



Université de Strasbourg
Faculty of Economics and
Management



Master Thesis:

A Methodological Pilot for Large-Scale Bibliometric Analysis of Artificial Intelligence in Science Using BigQuery

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**A thesis submitted in fulfillment of the requirements for the
degree of**

Master in Data Science for Economics and Business

Academic Year 2024-2025

Acknowledgements

I would like to express my deepest gratitude to my thesis supervisor, Professor Stefano BIANCHINI, for his guidance, valuable advice, availability, and constant support throughout this work. Having the opportunity to work under his supervision has been both a privilege and an enriching experience. His dedication, professionalism, and ability to create an academic environment that is both rigorous and welcoming made learning motivating and accessible.

My heartfelt thanks also go to my parents, Dr. Fidèle and Sylvane, for their unwavering support, patience, and constant encouragement. Their trust and sacrifices gave me the strength to carry this work through to completion. Without their presence, their guidance, and their love, this thesis would not have been possible.

I also wish to thank my sister, Dr. Délicia, as well as my brothers, Don and Joy, who have lightened this journey with their humor, complicity, and affection. Their encouragements and the moments we shared together were essential to maintaining my morale and moving forward with serenity.

I am sincerely grateful to my friend Brillant, a faithful companion since the very beginning of my undergraduate studies. Together, we faced challenges, celebrated achievements, and supported one another at every step of the way. His consistency, team spirit, and determination have been a constant source of inspiration and motivation to me.

I would also like to thank Sacha, my lifelong partner in study, with whom I had the privilege of sharing my master's years. His support, encouragement, and our unique complicity built on both seriousness and humor pushed me to give the best of myself. Thanks to him, I experienced the most memorable years of my studies, where hard work was always accomplished with joy and without ever feeling like a burden.

A very special mention goes to Carla, a true sister at heart, who has always been by my side through both good times and difficult ones. She has supported me in all my endeavors, lifted me up whenever things seemed to fall apart, and reminded me every single day that I was capable of moving forward and succeeding. Her trust, kindness, and encouragements have been an invaluable source of strength to me.

Finally, I extend my sincere thanks to all my friends, who, each in their own way, have offered me support, comfort, and encouragement. Their presence, their words, and their thoughtful gestures have made this academic journey lighter, more joyful, and truly unforgettable.

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Abstract

This thesis presents a methodological pilot study on how large-scale bibliometric data can be systematically leveraged to investigate the diffusion of artificial intelligence (AI) across scientific domains. Drawing on OpenAlex, a free and comprehensive bibliographic database and Google BigQuery, a scalable cloud-based data warehouse, we design a transparent and reproducible workflow intended as a foundation for future large-scale studies. The dataset covers publications from 2005–2024, organized around 181 level-1 scientific concepts selected to capture AI applications beyond core computer science. The analysis combines descriptive statistics and econometric models to assess two dimensions of research outcomes: scholarly impact, measured through citation-based indicators, and scientific creativity, captured through linguistic novelty metrics (semantic distance, new words, and novel phrase combinations). Results highlight strong and uneven growth in AI adoption across fields, with some disciplines showing rapid diffusion, while others remain marginal. To illustrate these heterogeneous patterns, the study integrates graphical dashboards of diffusion dynamics. These visual and analytical extensions underscore how AI acts as a transformative driver in certain areas, incremental in others and peripheral in a minority. The analyses are exploratory, aiming not to provide definitive empirical claims but to showcase the scalability and transparency of the proposed pipeline. In other words, this thesis contributes a reproducible methodological blueprint for mapping how AI reshapes knowledge production across science, while laying the groundwork for more extensive future investigations at the intersection of bibliometrics, data science and science policy.

1. Introduction

Science is the driving force behind many of our world's technological progress and economic development. From the Industrial Revolution to the age of the computer, scientific innovation has repeatedly revolutionized industries, encouraged productivity and profoundly improved the quality of life. But, even with this foundational role, recent developments have created anxieties about the present direction of scientific advancement. Several studies indicate that producing truly novel ideas has grown difficult and quite expensive. For example, [Bloom et al. \(2020\)](#) found that research productivity is sharply declining: it takes now far more effort and resources to produce the same volume of significant breakthroughs than just a few decades ago. This phenomenon could be attributed to the sheer scale of accumulated knowledge—the “knowledge circle” has become much broader, and it is that much more difficult for scientists to identify virgin areas or develop revolutionary approaches ([Chu et Evans, 2021](#)).

In this regard, Artificial Intelligence (AI) has already begun to revolutionize the conduct of science. AI is both a catalyst and an instrument that allows researchers to process vast amounts of data, reveal hidden patterns and even automate certain aspects of discovery that previously depended entirely on human intuition. Flagship examples, including DeepMind's AlphaFold2, illustrate the potential of AI: AlphaFold2 transformed structural biology by making predictions of protein structures with accuracy—a feat long regarded as an intractable problem for conventional computational techniques ([Jumper et al., 2021](#); [Griffin et al., 2024](#)). While such illustrations have attracted much attention, questions remain about how broadly AI is being integrated in research, which disciplines stand to benefit most, and whether adoption is equally distributed across scientific domains.

This thesis addresses these questions through an exploratory study illustrating the querying, processing, and analysis of large-scale bibliometric data systematically. While there is an emphasis on the method, generally speaking, the goal is to create a reproducible process that future research can continue to build upon in order to map AI diffusion and its effects on scholarly influence and scientific creativity more extensively. The study relies on OpenAlex, a big open-source bibliographic database, and uses Google BigQuery, a cloud platform supporting scalable analysis over large quantities of data, to build an accessible yet robust analytical pipeline.

This work is directly inspired by the study of [Bianchini et al. \(2025\)](#), who conducted a detailed bibliometric analysis of AI adoption across 80 concepts. While their approach offers depth and precision, our contribution emphasizes breadth and scalability by extending the coverage to 181 level-1 concepts. These concepts were purposively chosen as part of an attempt to exclude those that belong exclusively to the AI- or computer science-focused domain, in order to capture a wide range of potential application areas in which AI could diffuse and shape research work. Our analysis was conducted concept by concept, resulting in a dataset of more than 260 million records drawn from OpenAlex spanning 2005–2024.

Following this strategy, we present a descriptive statistical summary table and conduct illustrative econometric estimations of research influence and scientific creativity. Research influence is assessed through citation-based metrics, while scientific novelty is evaluated using recent novelty indicators developed in the bibliometrics literature. These analyses are not intended as exhaustive investigations, but rather as proof-of-concept demonstrations of the types of insights such an approach can generate.

The thesis is structured as follows. **Section 2** provides a literature review that situates the work within ongoing debates about stagnation in science, the transformative promise and risks of AI, and existing

empirical and theoretical contributions on AI adoption. **Section 3** describes the data sources and tools, namely OpenAlex and BigQuery and presents the workflow, showing step by step how bibliometric data were extracted and processed. **Section 4** presents descriptive statistics and exploratory analyses of AI diffusion, research impact and scientific creativity, with broad overviews and a case example of a field-level examination. **Section 5** reflects on the main challenges encountered, discusses potential applications of the proposed approach and outlines potential directions for future research. **Section 6** concludes.

2. Literature Review

2.1 Explanations for the Slowdown in Scientific Progress

Discussions about the future of science reveal a contradiction. Research systems are producing more than ever, but their ability to create major breakthroughs seems to be declining. Economic indicators, such as the productivity slowdown noted since the early 2000s ([Goldin et al., 2024](#)), reflect similar problems in research evaluation. Publications and patents are increasing, yet they are less likely to result in significant conceptual or technological advancements ([Scannell et al., 2012](#); [Miyagawa & Ishikawa, 2019](#); [Bloom et al., 2020](#); [Boeing & Hünermund, 2020](#)). Recent large-scale studies show that the potential for new work to change scientific direction, rather than just extend current paths, has dropped sharply ([Park et al., 2023](#)). As [Bloom et al., 2020](#) strikingly conclude, finding the ideas that drive exponential progress is becoming harder and harder.

Several explanations have emerged for this slowdown, each pointing to a different reason that limits the production of novelty. One interpretation highlights the natural dynamics of discovery. Historically, the first advances in a field focus on questions that are easiest to tackle and provide the biggest rewards. As these opportunities are exhausted, progress shifts to more complex and less rewarding areas. This viewpoint, often called the depletion of “low-hanging fruits,” suggests that the number of easily reachable insights is decreasing. This leaves future generations with tougher problems and smaller gains ([Cowen, 2011](#); [Gordon, 2017](#)).

Another perspective examines the growing demands of accumulated knowledge. Each new discovery adds to the body of material that aspiring researchers must understand before contributing. What once required expertise in a narrow field now needs years of specialized training. [Jones \(2009\)](#) described this as the burden of knowledge, arguing that as knowledge expands, so does the time and effort needed to create something original. Individual contributions are often replaced by collective efforts. Evidence supports this trend; landmark studies and patents are now largely produced by large teams, showing the division of cognitive tasks among specialists ([Wuchty et al., 2007](#)). The “death of the Renaissance man,” noted by [Jones \(2009\)](#), emphasizes that while the growth of knowledge is valuable, it also increases the barriers to innovation.

Alongside these structural factors, the way modern science is organized contributes its own inefficiencies. Time-consuming administrative tasks and competitive pressures leave less room for experimental or high-risk projects. Funding and publishing structures favor small advancements over major disruptions. Meanwhile, issues with replication and reliability have raised doubts about the strength of published findings, while inequalities in the academic system push for safer choices and lessen the diversity of perspectives ([Jones et al., 2008](#); [Stephan, 2012](#); [Azoulay & Li, 2020](#); [Franzoni et al., 2022](#)).

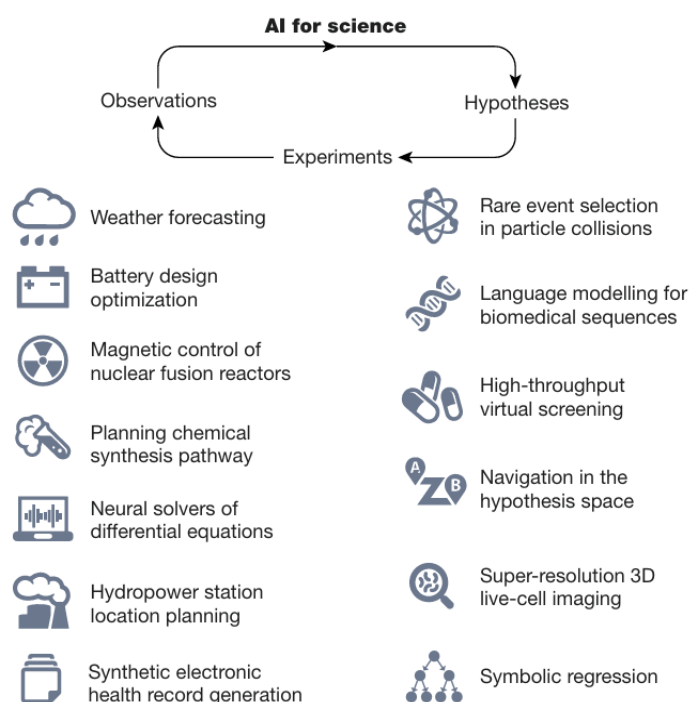
When considered together, these factors—the depletion of easy discoveries, the growing burden of accumulated knowledge, and institutional inflexibility—provide a complex view of why science seems

to be slowing down regarding disruptive potential. While [Paul Romer \(1993\)](#) once celebrated the nearly limitless search space for new ideas as the basis for ongoing growth, the current concern is that navigating this expanding landscape has become more resource-intensive and less likely to lead to unexpected discoveries.

This multidimensional slowdown sets the stage for discussions about the potential role of artificial intelligence (AI). Many see AI as a possible solution to each of these challenges. It can identify overlooked opportunities in vast datasets, help researchers integrate knowledge across fragmented domains, and automate repetitive or organizational tasks. This would give scientists more space for genuine discovery.

2.2 Science and the Rise of Artificial Intelligence

Figure 1: Overview of how artificial intelligence (AI) can contribute to scientific discovery, from data collection to hypothesis formation and theory building (source: [Wang et al., 2023](#)).



In contemporary discussions of the future of inquiry, artificial intelligence (AI) has come to be more and more characterized as not just a means but as a transformative technology with deep implications. Other writers have even referred to it as a universal approach to invention, arguing that AI can change science's mode of generating knowledge across a wide variety of domains ([Bianchini et al., 2022](#); [Cockburn et al., 2018](#)). The expectation is that AI can both increase productivity—by automating aspects of research—and enhance creativity, by opening up unexplored pathways that humans alone might not be able to see.

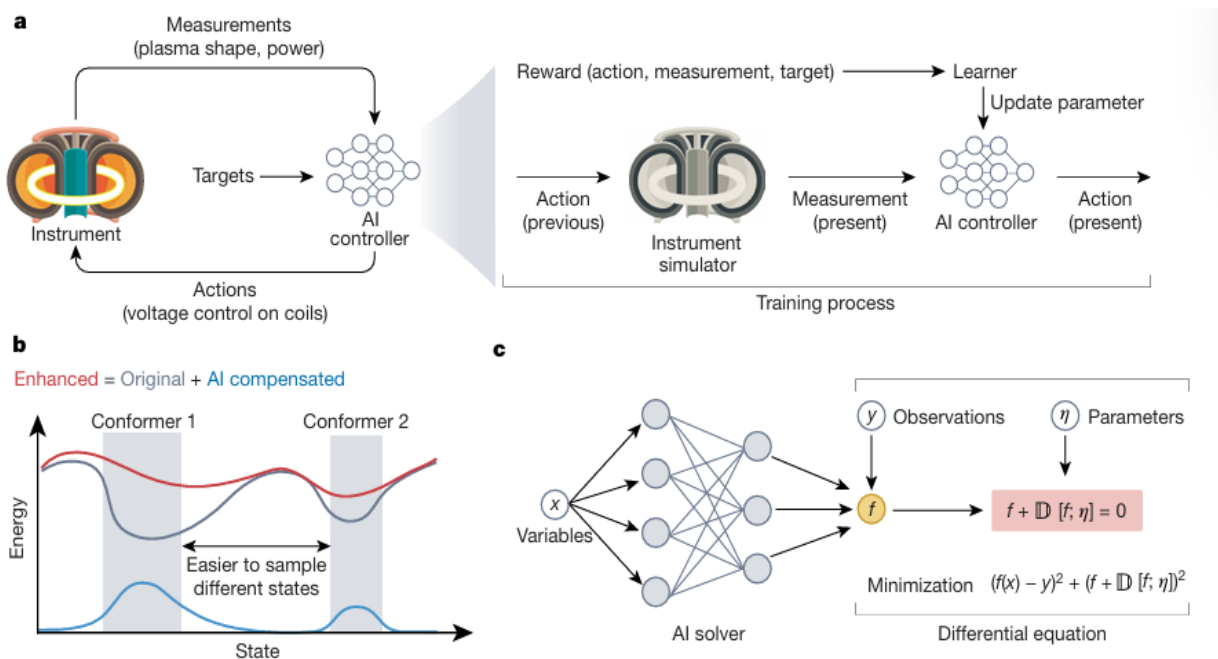
AI's potential is visible in the way it contributes at multiple stages of the research process. For example, it is capable of coordinating the organization of scientific work itself. Researchers have shown how intelligent systems can take on such simple "management" tasks in groups: task assignment, process guidance, merging output, shaping contributors, and enhancing learning through monitoring progress

and giving feedback (Koehler & Sauermann, 2024). Pilot studies illustrate this functionality in action—coordinating ecological surveys, such as for the mosquito population, through to overseeing observation campaigns of aurora events or motivating citizen scientists in large-scale projects (Carney et al., 2022; MacDonald et al. 2015; Segal et al., 2016). Artificial Intelligence does not displace scientists within these examples but acts as an intermediary node allowing collaborative research to be expedited.

Another way science is assisted by AI is through assisting scientists in resolving the staggering growth of knowledge ("knowledge burden") mentioned. AI offers practical comfort in the form of scanning and synthesizing enormous literatures, detecting unexplored themes, and making astounding connections between distant fields (Hastings, 2023). This role of information is supplemented by a more creative one: by recognizing unexpected patterns in simulations or data, AI may suggest new hypotheses. Neural networks can even be reversed to provide insight into representations they have learned, resulting in further conceptual investigation (Olah et al., 2017). Such applications illustrate why AI is already considered a collaborator for idea generation. Big projects like Microsoft's AI4Science are already applying these methods in areas ranging from drug discovery to materials science, structural biology, and mathematics (Microsoft Research AI4Science, 2023; Si et al., 2024).

Apart from hypothesis formation and literature, AI has also been compared to a computational microscope—a tool that increases the scientist's ability to measure and observe. It has been instrumental in an enormous array of applications: faster computer chip design (Mirhoseini et al., 2021) and cataloging genetic mutations (Karczewski et al., 2020). These successes demonstrate how AI can push the observational frontier further than that of traditional methods.

Figure 2: AI in scientific experiments and simulations.



Notes: Panel (a) shows AI control systems applied to nuclear fusion experiments, where an AI agent regulates magnetic fields based on real-time measurements to optimize performance. Panel (b) depicts the use of neural networks to accelerate molecular simulations by guiding the exploration of complex conformational spaces. Panel (c) presents physics-informed neural networks designed to solve partial differential equations by simultaneously estimating unknown parameters and fitting observed data. These examples collectively highlight AI's expanding role in enhancing experimental precision and computational modeling across scientific disciplines. (Source: Wang et al., 2023).

Farthest out on the edge of controversy is the possibility that one day AI would become an agent of understanding rather than a mere assistant. [Krenn et al. \(2023\)](#) describe what the conditions for such a scenario would be: an AI would need to foresee qualitative consequences from a theory without explicit computation, and then communicate that understanding to human scientists. Where this is speculative, it is the reason why some philosophers suggest rethinking whether computers will someday offer not just data, but concepts like theories (see also [De Regt, 2017](#)). Most would agree nowadays that current AI is very powerful at recognition and approximation, but has no real "scientific intelligence"—the ability to build causal explanations based on theory ([Mullainathan & Spiess, 2017](#); [Pearl & Mackenzie, 2018](#)).

Admiration of AI in science is tempered by severe threats. One of the primary concerns is its impact on novelty: because models are learned from past data, they are likely to reinforce current paradigms and not create new possibilities. This risks a paradigm lock that confines inquiry, along with a false sense of explanatory reach, where research appears wide but is actually constrained to what can be computed by computers ([Birhane et al., 2023](#); [Messerli & Crockett, 2024](#)). These procedures also threaten the role of serendipity, a process that previously has driven many breakthroughs.

No less pressing are transparency and trust issues. Complex models such as deep neural networks are usually "black boxes," producing outputs without clear explanations. As [Rudin \(2019\)](#) notes, approximations of uninterpretable predictions should not be mixed up with actual explanations. More importantly, risk is amplified by flawed or prejudiced training data, begetting the classic "garbage in, garbage out" issue ([Budach et al., 2022](#); [Hanson et al., 2023](#)).

Additional challenges are posed by recent generative models and large language models (LLMs). These can "hallucinate" coherent but false information ([Athaluri et al., 2023](#); [Beutel et al., 2023](#)), and their use generates plagiarism, overreliance on automated assistance, and compromising of research integrity. As [Birhane et al. \(2023\)](#) point out, ethical adoption of AI requires humility, skepticism, and commitment to scientific norms.

In summary, AI could potentially organize research, create novel ideas, increase experimental scope, and even propose theoretical impacts. Yet, it could also potentially encapsulate science, conceal reasoning, and compromise rigor if utilized irresponsibly. How seriously and responsibly the science community employs it will decide if it becomes a discovery accelerator or stagnation catalyst.

2.3 Empirical and Theoretical Evidence

Since artificial intelligence (AI) has grown rapidly in fields of research and nations, there have been several bibliometric studies charting its adoption ([Xu et al., 2021](#); [Gargiulo et al., 2023](#); [Hajkowicz et al., 2023](#); [Arranz et al., 2023](#); [AlShebli et al., 2024](#); [Schmallenbach et al., 2024](#); [Duede et al., 2024](#)). These reports consistently show AI application is now widespread and accelerating, yet their approaches reveal two persistent limitations: a tendency to focus on a select group of domains and technologies, and an excessive reliance on high-profile examples instead of overall trends, with the result that our understanding is partial.

The majority of the existing evidence depends upon research into landmark successes. For instance, research on the use of AI pattern recognition in the Kinect game system indicated improved knowledge generation in engineering fields ([Furman & Teodoridis, 2020](#)), while neural networks in health sciences were associated with higher citation rates compared to higher novelty ([Bianchini et al., 2022](#)). In structural biology, the arrival of AlphaFold created citations but not publication frequencies ([Yu, 2024](#)), demonstrating adoption is strong yet not uniform within scientific innovation. Furthermore, studies of

R&D in firms determine firms leveraging AI get improved innovation outcomes in patents as well as products (Rammer et al., 2022; Toner-Rodgers, 2024; Verendel, 2023).

An extremely informative piece of research by Gao and Wang (2024) goes beyond specific cases, examining long-term publication histories across a wide range of disciplines. Their evidence suggests that AI research accrues more citations—sometimes broadly—but that these effects are not uniformly spread, with some subdomains achieving greater advantages than others. Certain medical specialties, for example, perform better than broader categories, which is indicative of the influence of knowledge structure.

Despite these advances, several inherent constraints limit the field's advancement. Empirical research is individualistically focused on models or fields, leading to an emphasis on "success stories" rather than large-scale, replicable results. Creativity and novelty are rarely measured and compared systematically across countries or fields of knowledge; instead, the great majority of studies rely on common metrics like citations or patents, and as a consequence, issues of cross-field and geographic variability are underdeveloped.

Theoretical work complements this description by arguing that scientific progress depends on the ability to recreate existing knowledge into new configurations (Arthur, 2009; Weitzman, 1998; Fleming, 2001). Here, AI is commonly considered to be a means to transcend the weight of complexity: in that it assists scientists to get a better grip of huge combinatorial spaces more cost-effectively (Agrawal et al., 2018, 2024), and that it enables hybridization across domains as well as sheer mutation of ideas within them (Chen et al., 2024). This recombinant approach is especially relevant for understanding why impact varies by discipline: fields with fragmented, complex knowledge spaces benefit more from AI's search and integration capabilities, while smoother areas may see less dramatic effects.

When geographic trends are controlled for, it becomes obvious that countries differ significantly in the adoption of AI and in research output. While mapping these differences in adoption is well-established (Klinger et al., 2021; AlShebli et al., 2024; Schmallenbach et al., 2024), the study implies a gap in the measurement of the direct effect of AI on novelty and creativity at the country level.

All of these factors call for this approach: to move past slim samples, one technology or anecdote, and to scalable, systematic, reproducible methods. Literature today suggests it is hard to compare across fields and geographies in a reproducible way. It is only by constructing computational workflows that tap into open data infrastructures—such as OpenAlex and Google BigQuery—that one can work effectively with these challenges. This thesis therefore utilizes an analytical framework that is specifically designed to scale bibliometric analysis and, in the process, lay the groundwork for further, more extensive studies on the diffusion and effect of AI in science.

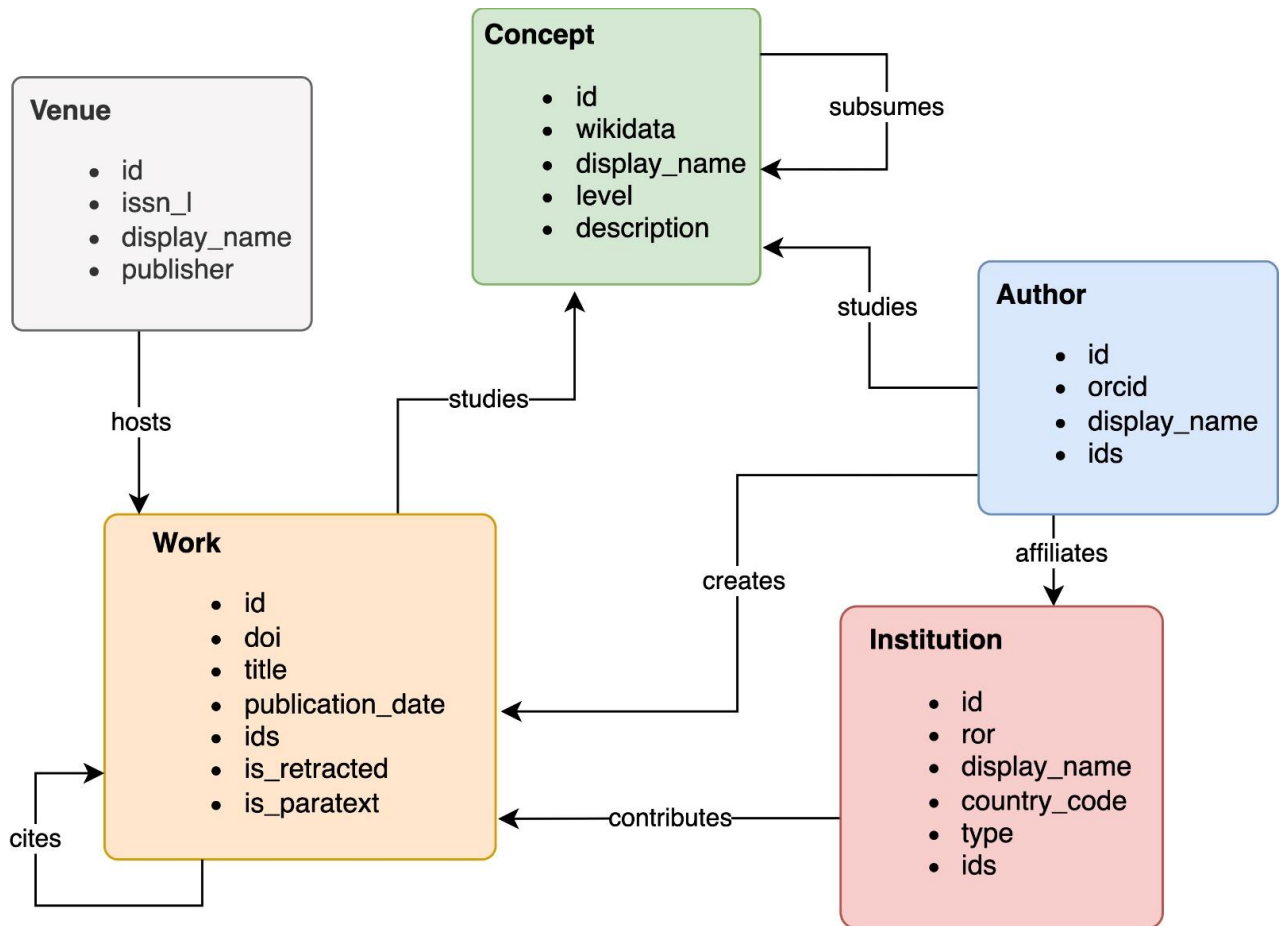
3. Methodology & Data Sources

3.1 OpenAlex: An Open Scholarly Universe

OpenAlex represents a major advancement in scholarly metadata availability and openness. Launched in 2022 as a successor to the proprietary Microsoft Academic Graph, OpenAlex was created with the goal of democratizing access to the entire global scientific record. By mid-2025, it had cataloged over 270 million scholarly works, encompassing a vast range of disciplines, publication types, and geographic regions. This coverage notably extends beyond the STEM fields to include social sciences and humanities, underscoring a commitment to a pluralistic and inclusive vision of science.

The data aggregated in OpenAlex originate from trusted repositories and indexing services such as CrossRef, ORCID, arXiv, and PubMed Central as well as institutional repositories and web crawls. This diverse sourcing ensures that users of OpenAlex enjoy near-complete visibility into global research outputs, complete with rich citation networks previously available primarily in commercial databases.

Figure 3: Representation of the main OpenAlex entities and their interactions



Structurally, OpenAlex models the scientific ecosystem through multiple interlinked entities. Central among these are:

- **Works**, which include journal articles, books, preprints, datasets, software, and conference papers. Each work is richly described through metadata fields like title, abstract, publication year, document type, citation counts, and references.
- **Authors**, uniquely identified individuals linked via persistent IDs such as ORCID, with recorded institutional affiliations spanning their careers.
- **Institutions**, comprising universities, research centers, companies, and other organizations tied to authors, enabling detailed geographic and organizational mapping of scientific activity.
- **Concepts**, which are topics assigned algorithmically to **Works**, arranged hierarchically from broad domains (e.g., “Computer Science”) to more specific subfields (e.g., “Machine Learning”). This layered classification allows for analytical flexibility: one can analyze broad disciplinary trends or zoom into precise technical areas.

Other entities include sources (journals, conferences), publishers, and funders. These components are not isolated; they form a complex network where works cite other works, authors collaborate, institutions affiliate with researchers, and funding flows between agencies and projects. This interconnectedness makes OpenAlex more than a database—it is a comprehensive map of global scholarship.

Among the many variables OpenAlex provides, key ones of particular analytical relevance include:

- Quantitative metrics like citation counts and author counts per work.
- Categorical metadata, including document types (article, preprint, dataset), open access labels, languages, and country or institution classifications.
- Relational identifiers used to maintain persistent links among entities, facilitating reproducibility and network construction.

OpenAlex’s public accessibility is notable—its full dataset is available free of charge both via a RESTful API, which supports complex, filtered queries, and as monthly-updated bulk data dumps for offline, large-scale analysis. Its open-source principles, comprehensive documentation and community-driven development make it a best practice model for reproducible bibliometric science.

Crucially for this thesis, OpenAlex’s unparalleled scale, depth of metadata and hierarchical topical classification perfectly support the goal of building a flexible and scalable framework to analyze the spread and impact of AI-related research across disciplines.

Given OpenAlex’s vast data coverage, direct access via the OpenAlex API might seem appealing. However, the API imposes limits—specifically, 100,000 requests per user per day and a maximum of 10 requests per second—that can significantly slow down large-scale data retrieval. Additionally, handling paginated responses and partial data can add complexity. These constraints make the API less suitable for extensive bibliometric analyses requiring millions of records. This motivated the use of cloud data warehouses such as BigQuery that complement OpenAlex’s openness with scalable analytic infrastructure.

3.2 Leveraging Google BigQuery: Scaling Up Bibliometric Analysis for Methodological Innovation

When analyzing large-scale bibliometric data, especially at the level demanded by this work, it quickly becomes apparent that traditional workflows struggle to keep pace with the complexity and volume of today’s scholarly record. This challenge led to the selection of Google BigQuery as the primary data retrieval engine for our project—a decision motivated by both the platform’s technical strengths and its alignment with open and extensible research practices.

Google BigQuery, launched in 2012, emerged from Google’s ambition to democratize data-intensive analysis. Drawing on innovations such as Dremel and Colossus, the platform was engineered to allow researchers to work directly with massive datasets in the cloud, thus eliminating the need for cumbersome local storage or traditional database management. This design choice ensures that, for the user, questions of server provisioning or parallel processing are essentially invisible, and the focus can remain squarely on research goals and clarity.

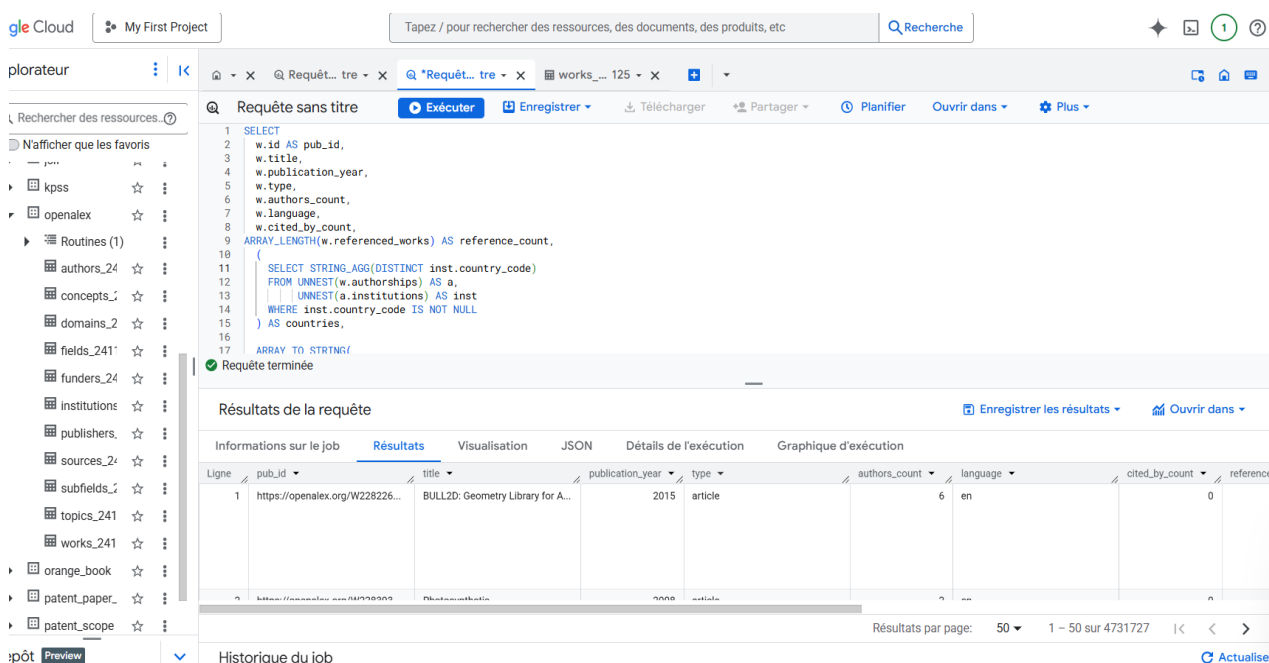
Flexibility and accessibility are at the core of BigQuery’s appeal. Researchers can interact with the platform through a user-friendly web console (see Figure 4), automate tasks using command-line tools, or integrate queries directly into coding environments such as Python or R. The use of the SQL querying

language not only makes it possible to construct complex data manipulations and aggregations with clarity and precision, but also facilitates sharing and transparency, as analytical code can be easily reviewed, reused, and adapted by others. Furthermore, BigQuery’s scalable architecture ensures that computational resources—both storage and processing power—expand automatically in response to a project’s demands, making it equally suited to pilot studies and long-term collaborative research.

Cost considerations are also crucial for methodological innovation, especially in academic contexts where resources are often limited. With BigQuery’s pay-for-what-you-use billing model, users are charged for the actual amount of data stored and processed by their queries. For those new to the platform, Google offers a \$300 free credit, valid for 90 days, which lowers the barrier to entry and enables extensive prototyping and experimentation before any expense is incurred.

Ordinarily, to work with external data in BigQuery, researchers would be required to upload large, structured files into the Google Cloud environment before analysis could begin. However, the workflow for this thesis benefited from a valuable exception: integration with the **Innovation Information Initiative** (I³)¹. The I³ project curates and maintains comprehensive open innovation data, including patent datasets, cross-domain citation graphs, funding records, and disambiguated author and institution identifiers—all organized in a dedicated BigQuery environment (the **nber-i3** project). Among these resources is a fully indexed, up-to-date version of the OpenAlex bibliometric dataset, made available for immediate cloud-native analysis.

Figure 4: Google BigQuery Interface.



Notes: The image shows the SQL editor (top) used to formulate queries and the resulting query outputs from the integrated OpenAlex dataset hosted in the nberI3 project (bottom). On the left, the interface displays the different tables available within the dataset.

This integration offered several pragmatic and technical advantages. Most importantly, it meant our analytical process could bypass the time-consuming step of uploading and structuring bulk data.

¹ For more information on the I³ project, see <https://iii.pubpub.org/>.

Instead, queries could be run directly on the pre-loaded OpenAlex dataset, leveraging not only its bibliometric depth but also the extended metadata and cross-dataset linkages curated by the I3 initiative. As a result, our team could prioritize the creation of reliable, modular queries rather than data engineering overhead.

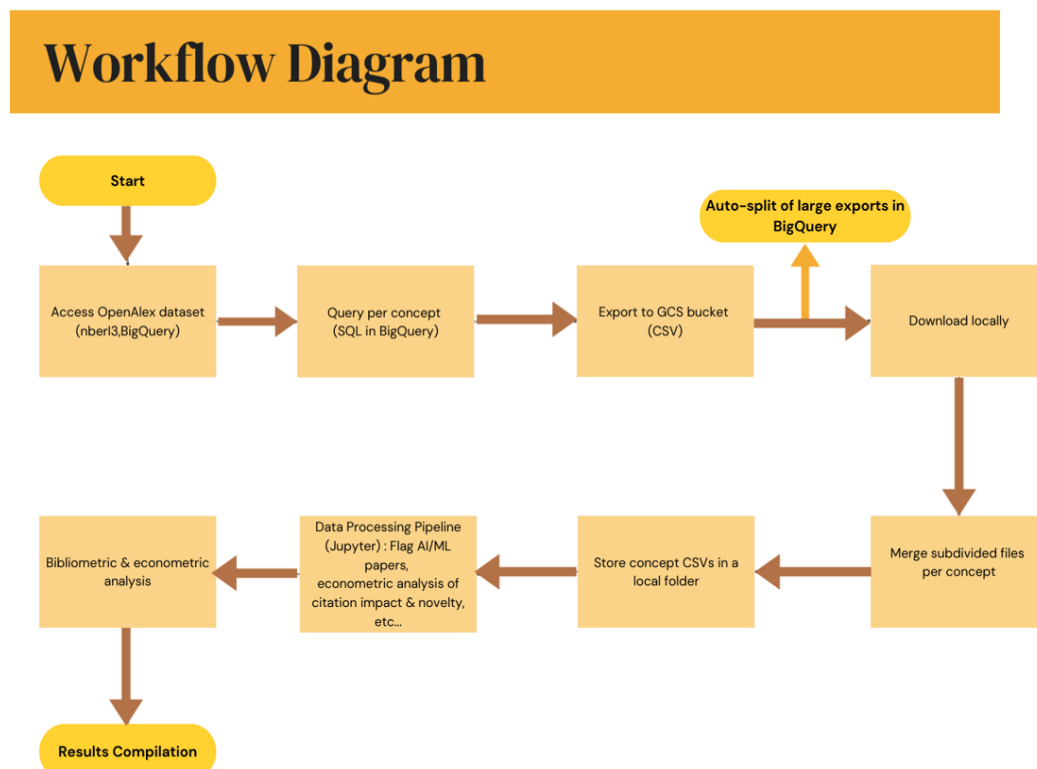
This approach presents a notable contrast to relying solely on the OpenAlex API. As noted earlier, the API is flexible and user-friendly for targeted queries, but its inherent rate limits and paginated outputs make it impractical for extracting or aggregating millions of publications at once. For large-scale bibliometric queries on the OpenAlex dataset, BigQuery can return results within seconds to minutes, even when handling tens of millions of publications. By contrast, using third-party libraries such as `openalex-api` in Python or `OpenAlex` in R—while flexible for targeted queries—would require hours or even days to retrieve comparable volumes of data, given the dataset’s scale.

In sum, by coupling OpenAlex’s exhaustively classified scholarly universe with the computational power and scalability of BigQuery—enabled by the foresighted integration work of the I3 initiative—this pilot demonstrates how cutting-edge digital infrastructure can transform bibliometric research from a laborious engineering task into an accessible, transparent, and collaborative scientific practice. The central use of SQL not only supports analytical rigor but also models best practices for transparent research workflows, ensuring that the strategies showcased in this thesis can be adopted, scrutinized, and extended by the broader scientific community.

3.3 Methodology: Operationalizing Data Access and Processing

Our analytical approach was mainly field-driven: we adopted a concept-by-concept strategy, selecting 181 fields from the OpenAlex database. This design (as shown in Figure 5) ensures broad disciplinary coverage, allowing comparisons and robust inferences across the full spectrum of scientific knowledge.

Figure 5: Overview of Data Workflow and Analysis Steps



We began by accessing the OpenAlex bibliometric database via Google BigQuery. For each concept, we engineered and executed detailed queries using SQL within this environment. These queries extracted granular metadata, including publication identifiers, titles, publication years, document types, language, citation metrics, reference counts and country affiliations. We applied precise filtering criteria to include only peer-reviewed articles and preprints published between 2005 and 2024 and written in English. Each publication record was additionally enriched with structured concept tags and institutional country codes, supporting subsequent analyses of disciplinary overlap and international collaboration.

To handle the large scale of data efficiently, the extracted tables for each concept were exported from BigQuery in CSV format to a secure Google Cloud Storage bucket. For particularly voluminous tables, BigQuery automatically subdivides the exports into multiple complementary files. We downloaded all files locally and subsequently merged subdivided CSVs per concept to create unified CSV per concept. They were then stored in a local folder to facilitate downstream processing.

For each scientific concept, we enhanced our dataset by linking the publications in our concept-specific files to a set of novelty indicators, drawn from “*Papers Textual Metrics*” CSV available on [Zenodo](#). These indicators capture dimensions of linguistic innovation, including the introduction of new words, phrases and combinations, their subsequent reuse in later literature, as well as measures of semantic distance from prior work.

Our data pipeline, developed in Python, was designed to clean, harmonize, and engineer variables essential for bibliometric and econometric analyses (see [Appendix for code](#)). Processing steps were conducted in the following sequence: first, citation velocity was calculated for each paper (citations per year since publication), and key variables such as author counts, reference counts, and citation metrics were standardized and log-transformed to manage skewed distributions. Next, we identified top-cited articles per year (top 10%, 5%, 1%) and flagged review-type papers.

A central step in the pipeline was the construction of an **AI/ML indicator variable** based on a comprehensive list² of AI and machine learning-related concepts curated from OpenAlex. This list included not only the core concepts of “Artificial Intelligence” and “Machine Learning,” but also a wide range of related subfields closely connected to AI/ML. Publications containing any of these concepts were marked as AI/ML-related, allowing us to calculate both absolute numbers and relative shares of AI-linked research across fields and over time. Finally, additional dummy variables captured international collaboration derived from the diversity of author institutional affiliations.

Following data preparation, we implemented two complementary econometric strategies to assess the role of AI in science. The first focused on bibliometric impact, using regression models to compare the citation performance of AI-linked and non-AI papers. The second addressed novelty, drawing on textual indicators such as new word and phrase introductions, combinations, and semantic distance. This dual framework allows us to evaluate AI’s influence on both scholarly visibility and research creativity.

All SQL queries, Python scripts, and data processing routines are documented and provided in the [appendices](#) to ensure full transparency and reproducibility.

² The list of concepts used to flag AI papers was developed by taking a small set of core AI fields (Artificial intelligence, Machine learning, Computer vision, Speech recognition, and Natural language processing) as a starting point. From there, related methods were added when they were clearly linked to these core areas, while unrelated or overly general concepts were removed. This process helped create a list that is broad enough to capture the diversity of AI research, but focused enough to avoid noise from non-AI fields.

4. Results: Patterns of AI Diffusion and Impact Across Fields

4.1 Cross-Disciplinary Overview of AI Diffusion and Impact

This section presents the key results and findings obtained through the application of the previously described framework. A comprehensive results table was constructed, covering descriptive indicators, econometric estimates of impact and econometric estimates of novelty. To facilitate interpretation, excerpts are presented separately in the subsections below. The complete table is available upon request.

4.1.1 Descriptive Statistics on AI Diffusion

We assembled a results table spanning 181 conceptual domains between 2005 and 2024 (see **Table 1**). For each concept, the table reports:

- **Total number of publications,**
- **The count of AI-linked publications both overall and by year,**
- **And the share of AI publications both overall and by year.**

Table 1: Descriptive statistics of AI diffusion across some of the 181 scientific fields (excerpt)

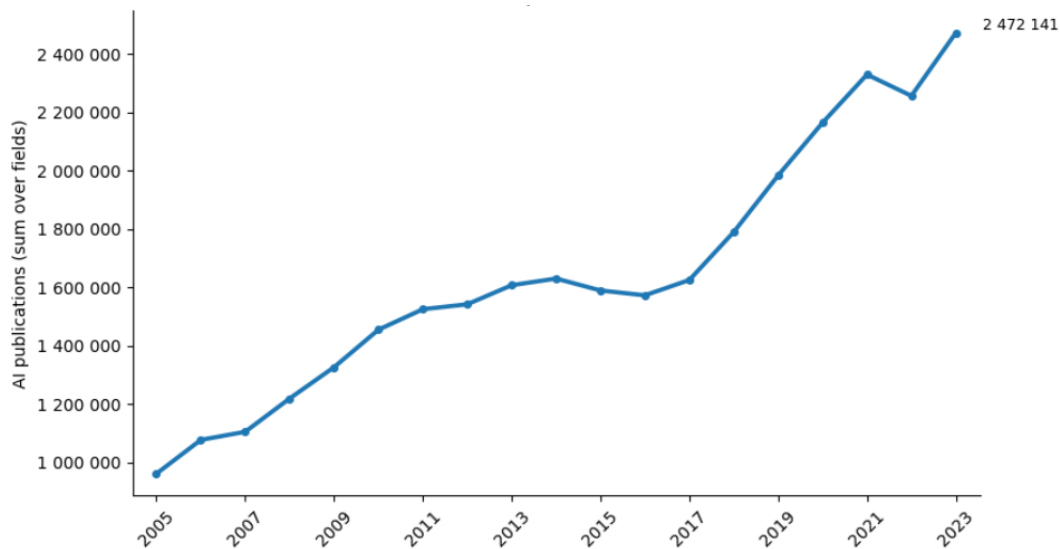
Concept	No. of Pub.	No. of AI Pub.	Share of AI papers	No. of Publ. in 2005	No. of AI Pub. in 2005	...	No. of Pub. in 2024	No. of AI Pub. in 2024
Astrobiology	466 178	15 582	3.34%	17 487	511	...	20 839	941
Cardiology	1 587 237	56 286	3.55%	56 603	1 768	...	93 584	3980
Computational Chemistry	458 068	21 916	4.78%	15 284	563	...	22 333	1 508
Construction Engineering	136 394	20 326	14.9%	5 777	776	...	6 182	804
Development Economics	357 753	15 520	4.34%	10 985	426	...	13 417	446
Environmental Resource Management	664 203	67 535	10.17%	17 470	1 845	...	41 913	3 922
Quantum Electrodynamics	127 270	4 943	3.88%	5 740	211	...	7 923	275
Linguistics	5 391 248	1 286 197	23.86%	154 065	31 373	...	221 738	64 802
Neuroscience	2 813 925	552 165	19.62%	78 558	13 630	...	157 981	30 349
Polymer Science	127 213	2 716	2.14%	5 747	94	...	11 121	315

Note: This excerpt shows a subset of fields and years selected for illustrative purposes. The full table, including all 181 fields and the complete time series, is available upon request.

This excerpt underscores the wide variation in the pace and extent of AI adoption across disciplines. For example, while linguistics and neuroscience already display relatively high AI shares, fields such as polymer science or quantum electrodynamics remain at lower levels of adoption.

To contextualize these patterns, **Figure 6** presents trends in the aggregate number of AI-linked publications per year from 2005 to 2023.

Figure 6: Total number of AI-linked publications (2005-2023)

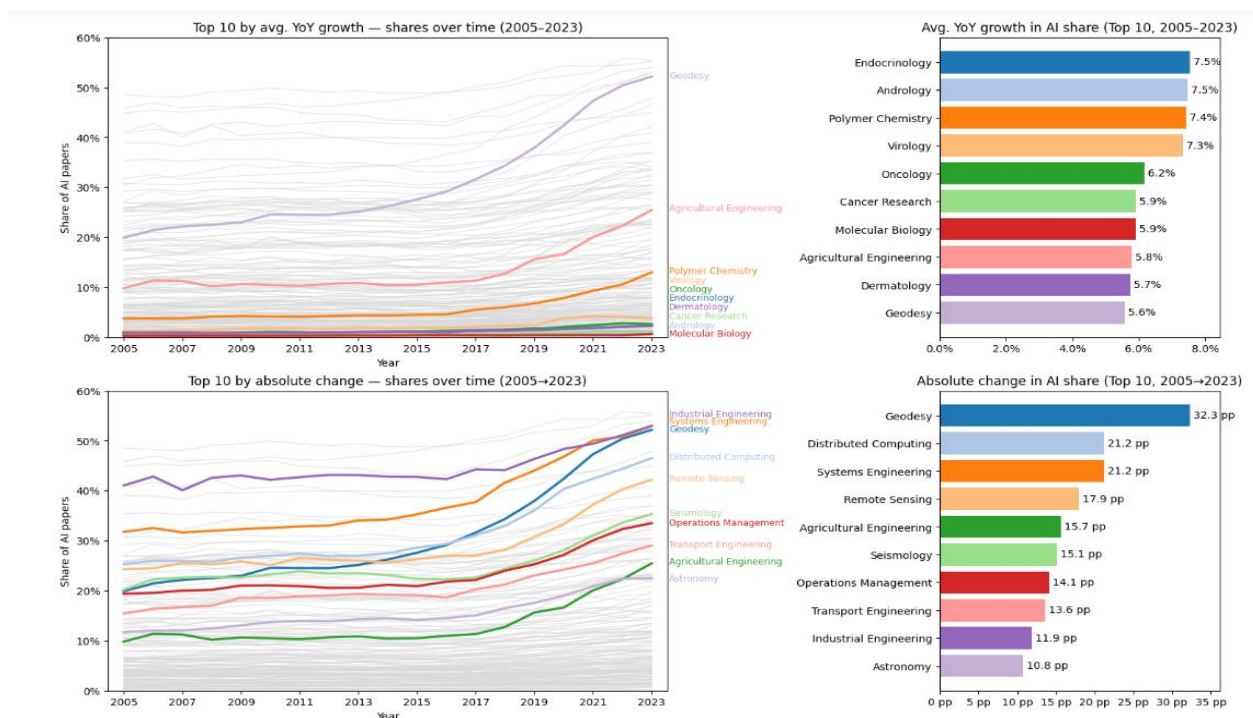


Note: This line plot displays the aggregate annual count of AI-attributed publications across all studied fields, illustrating the overall acceleration of AI research activity.

The trajectory reveals a strong and nearly uninterrupted increase, with a sharp acceleration after 2017. By 2023, AI-linked outputs surpassed two million publications, confirming the rapid and transversal diffusion of AI within the global scientific system.

Beyond aggregate growth, it is important to distinguish between disciplines that have experienced the fastest relative increases and those that have undergone the largest absolute shifts. **Figure 7** presents a comparative dashboard summarizing both the average year-on-year (YoY) growth and absolute share increase for all studied concepts, ranking the top 10 fields in each dimension.

Figure 7: Year-on-year (YoY) growth and absolute growth in AI publication shares for our list of concepts, 2005–2023



Note: **Left panels** show temporal trajectories of the fastest-accelerating and largest-gaining fields (top 10); **right panels** provide direct rankings, highlighting both relative “hotspots” and domains with the greatest cumulative transformation.

The results show two distinct patterns. Some fields, like endocrinology, andrology, polymer chemistry, and oncology, have high year-on-year growth rates, though starting from relatively small bases. In contrast, areas such as geodesy, systems engineering, and remote sensing have seen the largest absolute increases in AI share, reflecting more substantial changes in their publication portfolios.

Together, these descriptive perspectives benchmark the evolving status of AI across science and provide a foundation for subsequent analyses of research influence (citation metrics) and scientific creativity (novelty)

While these descriptive statistics highlight uneven adoption dynamics, they do not reveal whether AI-linked publications differ in terms of influence or novelty. To address this, we turn to econometric models at the field level.

4.1.2 Econometric Models of Impact

First, we estimated four econometric models for each field to assess whether AI/ML-linked papers differ systematically in its citation outcomes.

- **OLS model on impact** assesses whether AI papers tend to receive more citations per year than comparable non-AI papers.

Formally:

$$tc_w_log = \beta_0 + \beta_1 is_aiml + \beta_2 author_count_log + \beta_3 reference_count_log + \beta_4 intern_collab + \beta_5 survey + \sum_{2005}^{2024} \gamma_t D_{year} \quad (1)$$

where the dependent variable is the log of weighted citation^{*} counts per year. The key parameter is the AI/ML dummy (1 if the paper is AI-related). Controls include log number of authors, log number of references, international collaboration³, and survey status⁴. Year fixed effects are included.

^{*}Weighted Number of Citations (*tc_w*) is the total citations a paper has received up to 2024 (the time of data extraction), divided by the number of years since its publication. This gives an annualized citation rate that makes papers of different ages comparable.

- **Logistic models on top citation outcomes:** We then estimate whether AI/ML-linked papers are more likely to be among the most cited publications in their field. The dependent variable is a binary indicator equal to 1 if the paper belongs to the top 10%, 5%, or 1% of most cited papers in the same field and year.

Expressed as:

³ International Collaboration is coded as 1 when a paper has authors from two or more different countries, and 0 otherwise.

⁴ The Survey/Review indicator is set to 1 if the paper’s title contains the words ‘survey’, ‘review’, or ‘overview’, and 0 if not.

$$Pr(x_{most\ cited}) = \text{logit}^1(\beta_0 + \beta_1 is_aiml + \beta_2 author_count_log + \beta_3 reference_count_log + \beta_4 intern_collab + \beta_5 survey + \sum_{2005}^{2024} \gamma_t D_{year} \quad (2)$$

where $x \in \{10, 5, 1\}$.

The independent and control variables are identical to the OLS specification.

Table 2: Econometric estimates of citation outcomes for AI vs. non-AI papers (excerpt)

Concept	OLS Coeff.	OLS (SE)	Logit 10 %	Logit 10 % (SE)	Logit 5 %	Logit 5 % (SE)	Logit 1 %	Logit 1 % (SE)
Astrobiology	-0.043***	0.003	-0.187***	0.036	-0.192***	0.050	-0.113	0.105
Cardiology	0.026***	0.002	0.394***	0.016	0.510***	0.020	0.617***	0.041
Computational Chemistry	0.078***	0.004	0.380***	0.023	0.490***	0.030	0.792***	0.055
Construction Engineering	0.017***	0.003	0.039	0.030	0.078**	0.038	-0.015	0.075
Development Economics	0.011***	0.004	0.044	0.028	-0.014	0.037	-0.079	0.079
Environmental Resource Management	-0.007***	0.002	-0.035**	0.015	-0.040**	0.019	-0.077*	0.040
Quantum Electrodynamics	-0.018**	0.008	-0.187***	0.055	-0.243***	0.076	-0.068	0.153
Linguistics	0.022***	0.000	0.135***	0.004	0.264***	0.005	0.558***	0.009
Neuroscience	-0.008***	0.001	0.165***	0.006	0.261***	0.007	0.468***	0.015
Polymer Science	0.030***	0.010	0.130*	0.074	0.197**	0.091	0.480***	0.158

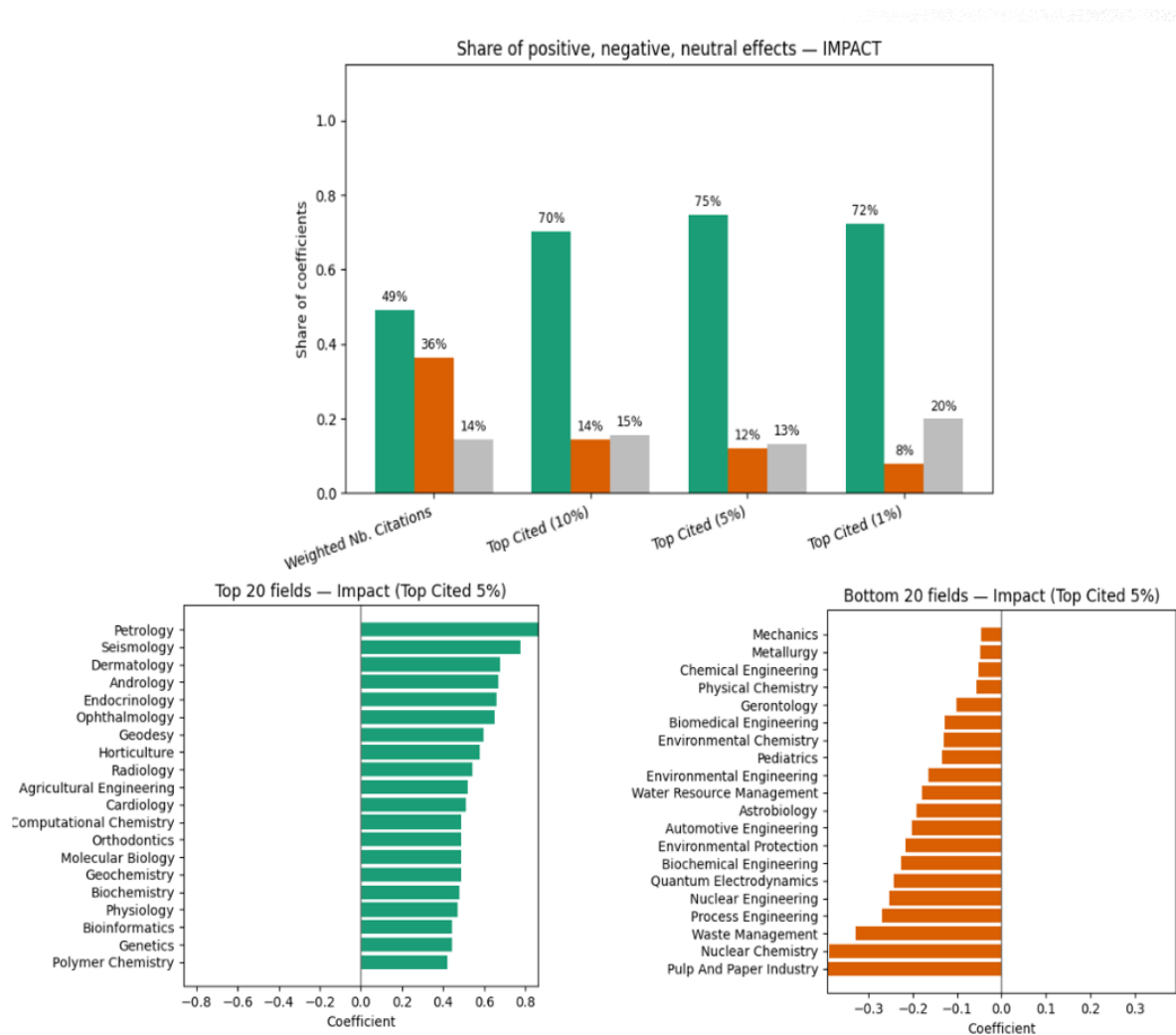
Note: This excerpt shows a subset of fields and years selected for illustrative purposes.

*Significance levels are indicated with *, **, and *** for $p < 0.1$, 0.05 , and 0.01 , respectively. The full results, for all 181 fields are available upon request.*

Table 2 presents excerpts of the estimated coefficients and robust standard errors for a subset of fields for illustrative purposes. These results quantify whether AI-linked papers tend to accumulate more citations, reflecting greater scientific impact and whether they are disproportionately represented among the most highly cited publications.

To provide a clearer visual overview, **Figure 8** summarizes the estimated citation coefficients across all 181 fields, highlighting both average effects and the most/least affected fields.

Figure 8: Impact coefficients for AI vs. non-AI papers across fields.



*Note: **Top panel** shows the distribution of positive, negative and neutral coefficients from regression models estimating the association between AI adoption and citation outcomes (weighted number of citations, and probability of being in the top 10%, 5%, or 1%). **Bottom panels:** top 20 (left) and bottom 20 (right) fields ranked by coefficient size for the “Top Cited 5%” specification.*

The figure shows that AI-related publications perform better in terms of scholarly impact: around 70–75% of coefficients are positive when looking at highly cited papers (top 10%, 5%, 1%) with only a small minority of negative and 14–20% neutral. However, disciplinary heterogeneity is pronounced. Fields such as petrology, seismology, dermatology, and cardiology are among the strongest positive outliers, where AI-linked work has a significantly higher probability of being among the most cited. By contrast, disciplines like chemical engineering, process engineering, nuclear chemistry, and waste management display negative or near-zero coefficients, suggesting little to no citation advantage from AI adoption.

Thus, these results suggest that the diffusion of AI across science does not yield uniform gains in scholarly recognition. In many fields—particularly those dealing with complex data, diagnostics, or modeling—AI appears to enhance visibility and impact, likely because it enables methodological breakthroughs and novel insights. However, in other areas such as traditional engineering or applied chemistry, AI adoption seems less transformative, possibly because these domains rely more on

established experimental protocols and incremental improvements. While indicative, these results reflect only the concepts included in this analysis and may not capture broader trends across science.

4.1.3 Econometric Models of Novelty

Complementing citation analysis, we examine whether AI/ML papers are associated with distinct novelty patterns.

Five econometric models were estimated:

- **OLS model on semantic distance** quantifies how “novel” a paper is in terms of the conceptual combinations of its words and phrases compared to the existing literature.

$$\text{semantic_distance} = \beta_0 + \beta_1 \text{is_aiml} + \beta_2 \text{author_count_log} + \beta_3 \text{reference_count_log} + \beta_4 \text{intern_collab} + \beta_5 \text{survey} + \sum_{2005}^{2024} \gamma_t D_{\text{year}} \quad (3)$$

- **Logistic models** are estimated for the following binary novelty indicators⁵:
 - *new_word_d*: A binary indicator set to 1 when a paper introduces at least one new single word (unigram) in its title or abstract that is later adopted by at least one other publication.
 - *new_word_comb_d*: A binary indicator set to 1 when a paper introduces at least one novel combination of two words, with the requirement that this exact pair is reused in at least one subsequent publication. The individual words themselves do not need to be new.
 - *new_phrase_d*: A binary indicator set to 1 if a paper introduces at least one new noun phrase—defined as a group of one or more words centered on a noun, such as “gene expression regulation”—in its title or abstract, which is later reused at least once.
 - *new_phrase_comb_d*: A binary indicator set to 1 when a paper introduces a new pairing of phrases used together for the first time and this phrase combination is subsequently cited in at least one other publication.

These indicators capture not only whether a linguistic innovation appeared but also whether it was **subsequently reused in later publications**, signaling **enduring rather than ephemeral novelty**.

Let Y_i denote any of the four binary novelty outcomes (*new_word_d*, *new_word_comb_d*, *new_phrase_d*, *new_phrase_comb_d*).

The model is:

$$\Pr(Y_i = 1) = \text{logit}^1(\beta_0 + \beta_1 \text{is_aiml} + \beta_2 \text{author_count_log} + \beta_3 \text{reference_count_log} + \beta_4 \text{intern_collab} + \beta_5 \text{survey} + \sum_{2005}^{2024} \gamma_t D_{\text{year}}) \quad (4)$$

⁵ We sourced the initial novelty metrics from [here](#) (Art et al. 2025) and linked them to our dataset using the OpenAlex publication ID. All variables—aside from semantic distance—were developed by [Bianchini et al. \(2025\)](#) specifically for this analysis.

Where:

- $Y_i = 1$ if the paper introduces a reusable innovation (word, word combination, phrase, or phrase combination)

All models include the same controls as the citation regressions: log number of authors, log number of references, international collaboration, survey status and year fixed effects.

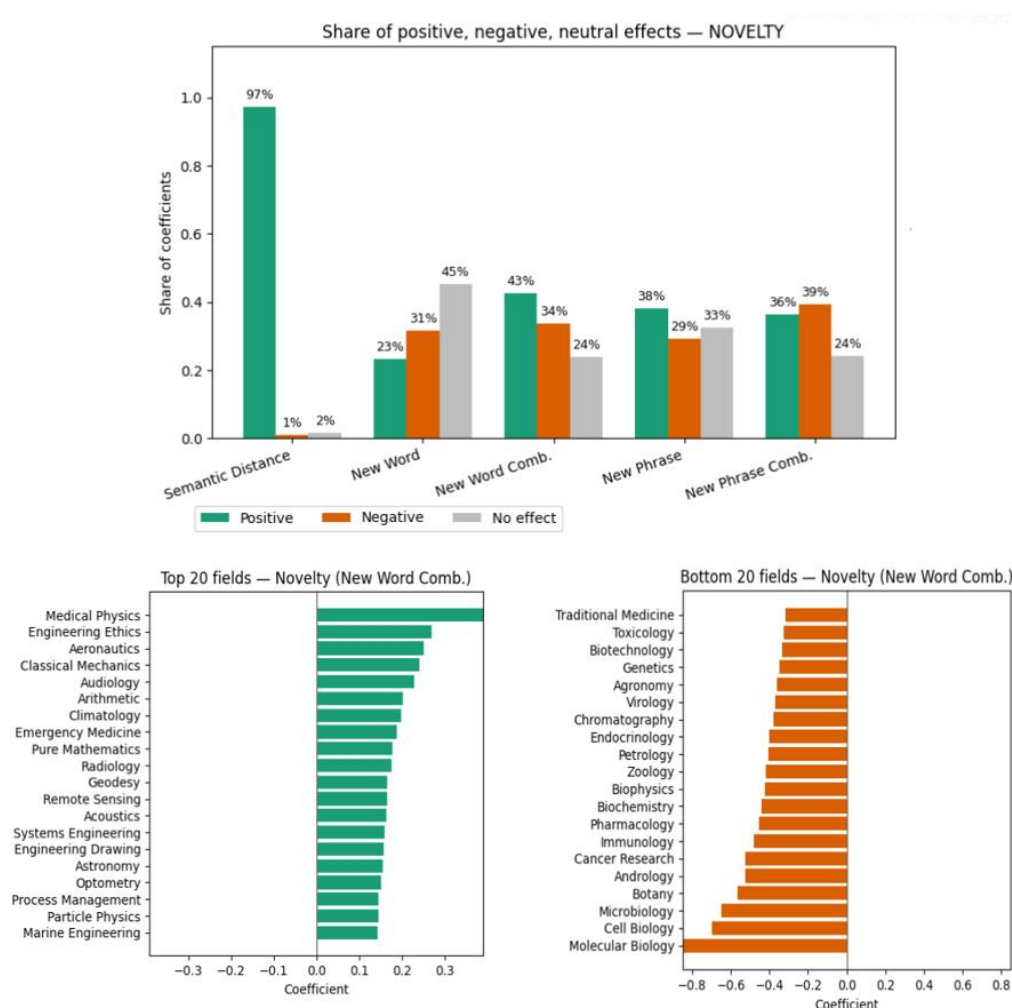
Table 3: Econometric estimates and standard errors of novelty outcomes for AI vs. non-AI papers (excerpt)

Concept	OLS Semantic Distance Coef (SE)	Logit New Word Coef (SE)	Logit New Word Coef (SE)	Logit New Phrase Coef (SE)	Logit New Phrase Comb Coef (SE)
Astrobiology	0.030*** (0.001)	-0.009 (0.076)	0.081** (0.033)	0.099*** (0.038)	0.007 (0.035)
Cardiology	0.022*** (0.000)	0.152*** (0.029)	0.137*** (0.013)	0.299*** (0.015)	0.012 (0.014)
Computational Chemistry	0.028*** (0.000)	NaN	NaN	NaN	NaN
Construction Engineering	0.005*** (0.001)	0.008 (0.080)	NaN	NaN	NaN
Development Economics	0.006*** (0.000)	-0.265*** (0.097)	-0.112*** (0.026)	-0.028* (0.016)	-0.051*** (0.012)
Environmental Resource Management	0.007*** (0.000)	0.006 (0.032)	0.037*** (0.012)	-0.028* (0.016)	-0.051*** (0.012)
Quantum Electrodynamics	0.005*** (0.001)	NaN	NaN	NaN	NaN
Linguistics	0.015*** (0.000)	0.100*** (0.008)	0.090*** (0.003)	0.154*** (0.004)	0.082*** (0.004)
Neuroscience	0.015*** (0.000)	0.100*** (0.008)	0.090*** (0.005)	0.046*** (0.005)	0.053*** (0.005)
Polymer Science	0.016*** (0.001)	NaN	NaN	NaN	NaN

Note: For a subset of fields, logistic models on novelty outcomes failed to converge, resulting in missing estimates (NaN). This typically occurs when the dependent variable exhibits little or no variation in certain years (e.g., all observations equal to 0 or 1), leading to singularity problems in the variance–covariance matrix.

To synthesize these results, **Figure 9** presents the distribution of coefficients across novelty measures as well as the 20 most positive and 20 most negative concept-level associations for the “new word combination” specification.

Figure 9: Novelty coefficients for AI vs. non-AI papers across fields.



*Notes: **Top panel** displays the distribution of positive, negative, and neutral coefficients from regression models estimating the association between AI adoption and different novelty measures (semantic distance, new words, new word combinations, new phrases, and new phrase combinations). **Bottom panels:** top 20 (left) and bottom 20 (right) fields ranked by coefficient size for the “New Word Combinations” specification.*

A dual pattern is revealed. On one hand, semantic distance is almost universally positive: 97% of coefficients show that AI papers connect more distant areas of knowledge. In other words, AI-linked papers often combine ideas from fields that usually do not overlap, creating novel connections across scientific domains. On the other hand, textual novelty indicators are more heterogeneous: Roughly one third of coefficients are positive, one third negative and one third neutral, with notable variation across measures. Disciplines such as medical physics, engineering ethics, aeronautics, and climatology are among those with the strongest positive associations with AI in terms of generating new word combinations. In contrast, molecular biology, cell biology, microbiology, and cancer research are among the most negative outliers, suggesting that AI adoption in these fields does not systematically generate new linguistic or textual markers of novelty.

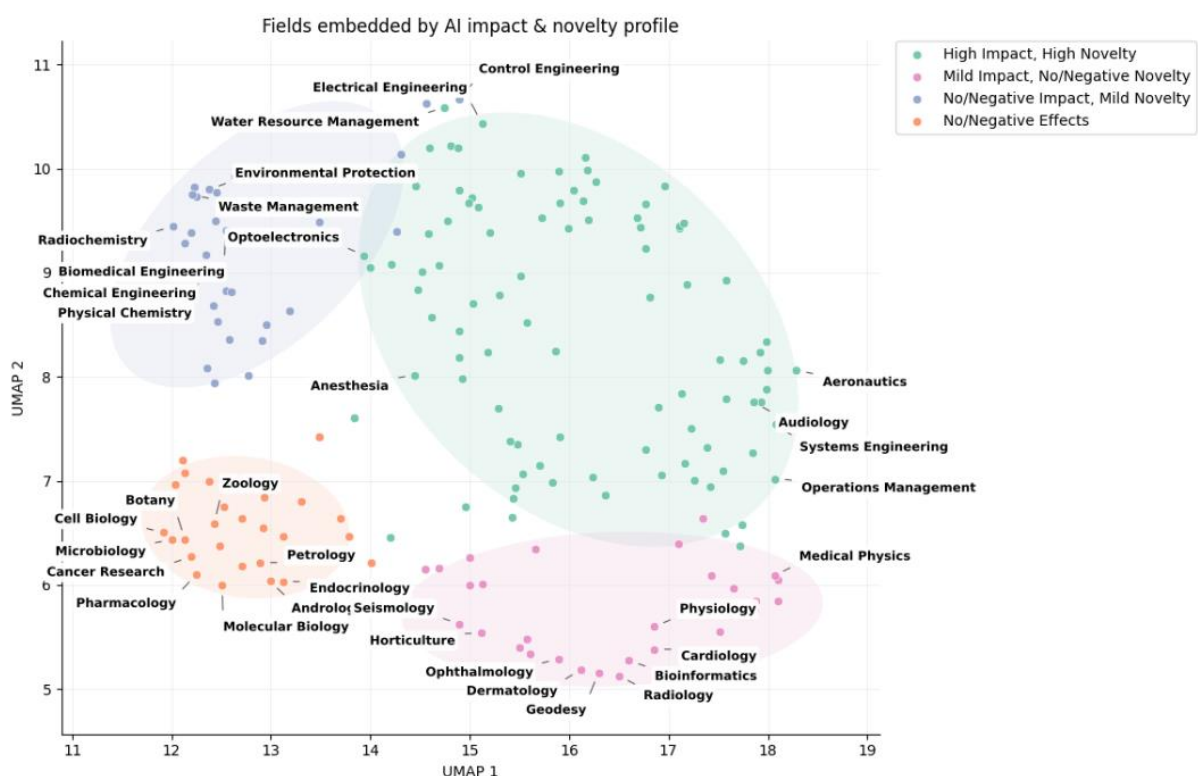
These findings suggest that the diffusion of AI across science contributes more to **conceptual recombination** than to **linguistic innovation**. AI seems to expand the semantic space by connecting

previously distant areas of knowledge, but it does not consistently produce new terminology or textual markers of novelty.

This pattern may reflect how AI is integrated: in technical or interdisciplinary fields (e.g., engineering, physics, climatology), AI enables new problem framings and recombinations, resulting in measurable novelty. By contrast, in highly standardized fields such as molecular biology or cell biology, AI may reinforce established research vocabularies rather than introduce new ones. It is important to note that these observations are based on the 181 concepts analyzed in this study and should not be interpreted as fully representative of the entire scientific landscape.

To explore patterns across all these concepts simultaneously, we applied clustering methods to position them according to their AI-related impact and novelty profiles, providing a system-wide view of AI diffusion across science.

Figure 10: Clustering of 181 scientific fields by AI-related impact and novelty



Note: Each point represents a scientific field. Clusters are formed based on similarities in AI adoption, citation impact, and novelty outcomes.

The analysis reveals four broad clusters:

- **High impact, high novelty (Green cluster):** This group includes engineering and applied sciences such as control engineering, electrical engineering, systems engineering, aeronautics, operations management, and environmental protection. In these domains, AI contributes simultaneously to conceptual novelty and scholarly impact, which is coherent given their proximity to AI's technical core and the availability of structured problems and abundant data.

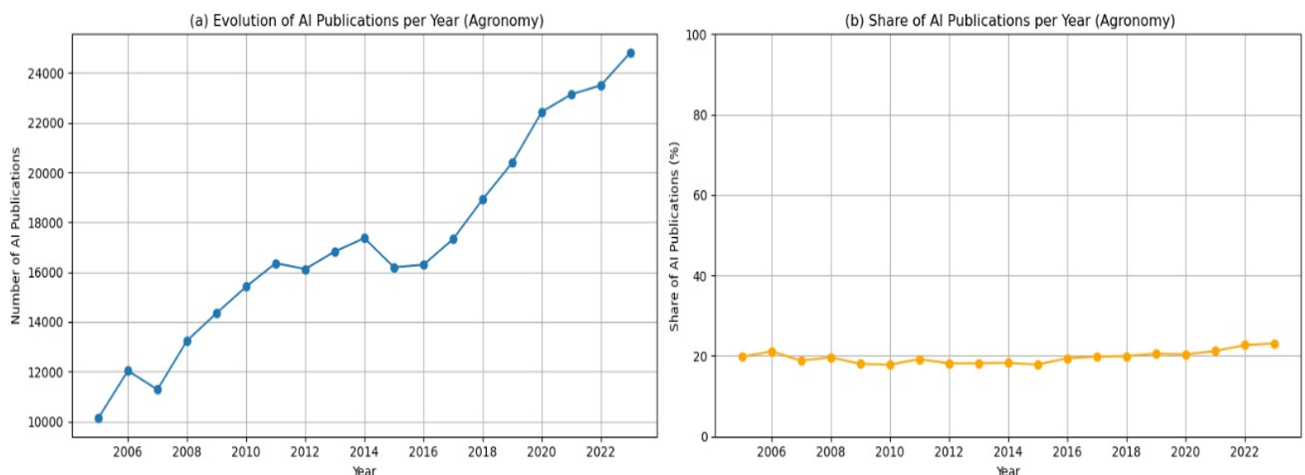
- Mild impact, no/negative novelty (**pink cluster**): Biomedical and medical-adjacent fields such as medical physics, cardiology, radiology, physiology, and ophthalmology belong here. AI enhances impact, mainly through improvements in diagnostics and imaging, but it does not substantially reshape conceptual paradigms.
- No/negative impact, mild novelty (**blue cluster**): Chemistry-related fields, biomedical engineering, optoelectronics, and radiochemistry display novelty gains through new methods and explorations, yet these advances have not yet translated into higher scholarly recognition.
- No/negative effects (**orange cluster**): Disciplines such as zoology, botany, cell biology, microbiology, and cancer research show neither significant impact nor novelty effects from AI. In these domains, integration remains limited, likely due to the complexity of experimental data and the incremental nature of AI applications.

This clustering underscores the heterogeneous nature of AI's diffusion across the 181 scientific fields included in the study. In some domains, AI acts as a transformative driver; in others, it provides incremental improvements; and in a minority, its role remains peripheral. The analytical structure developed in this thesis thus enables systematic mapping of disciplinary differences in how AI reshapes knowledge production.

4.2 Concept-Level Analysis: Agronomy Case Study

To complement the aggregate perspective, we present a field-level example study in Agronomy, illustrating how AI diffusion can be tracked over time and across major contributing regions (China, EU27, USA). This example is intended for illustrative purposes and should not be interpreted as representative of the broader scientific landscape.

Figure 11: Trend in the total number and share of AI papers in Agronomy (2005-2024)



Notes: Panel (a) illustrates the total volume of AI-related publications produced over time in Agronomy. Panel (b) presents the share of AI papers as a percentage of total output within the field.

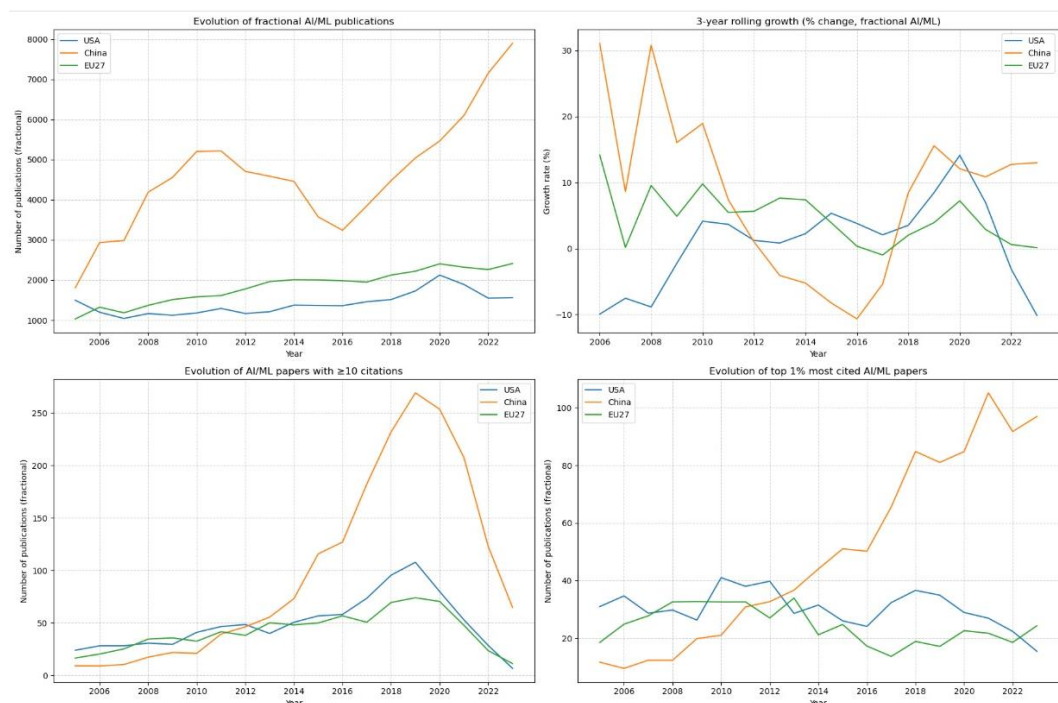
Panel (a) shows the absolute evolution of AI-related publications in Agronomy between 2005 and 2023. The series begins with around 10,000 publications in 2005, rising steadily to reach more than 24,000 by 2023. The first decade (2005–2015) is marked by incremental but consistent growth, maybe reflecting the gradual integration of computational techniques into agricultural research. After 2016, the curve steepens, pointing to an acceleration in AI adoption within Agronomy. This shift could

correspond to a phase of technological diffusion: as AI tools became cheaper (via open-source frameworks and cloud computing) and data availability expanded (remote sensing, IoT in farming, genomic datasets), the marginal cost of applying AI fell sharply, encouraging broader uptake. At the same time, demand-side pressures—linked to food security, climate adaptation, and productivity challenges—could have acted as a pull factor, increasing the value of AI applications in crop monitoring, yield prediction and resource optimization. The strong post-2016 growth thus reflects both supply-side enablers and demand-driven incentives.

Panel (b) presents the share of AI papers as a proportion of all agronomy publications. Despite the sharp rise in absolute numbers, the relative share remains remarkably stable, fluctuating between 18% and 22% for most of the period, and only climbing modestly to 23% by 2023. This stability implies that the rest of agronomy—non-AI research streams such as soil sciences, plant breeding, and climate studies—also expanded substantially. This suggests that AI has functioned more as a complementary innovation rather than a displacing one. Research diversification persists: while AI lowers costs and increases efficiency in certain tasks, other areas of agronomy continue to attract investment and generate knowledge, keeping the field balanced. Barriers such as the need for domain-specific data, may have limited AI expertise in agronomy labs, and high costs of validation could also help explain why AI’s share has not surged more aggressively.

The two panels highlight a classic general-purpose technology pattern: rapid expansion in absolute adoption (extensive margin) without a proportional surge in dominance (intensive margin). Agronomy is not becoming “AI-only,” but rather experiencing a broad-based expansion where AI grows in parallel with other scientific approaches. The modest upward drift in the AI share after 2020 may foreshadow a gradual intensification of AI’s role as adoption barriers continue to fall.

Figure 12: AI research output across key regions in Agronomy (USA, China, EU27)



Notes: Panel (a) shows the fractional⁶ share of AI-related publications over time in China, the EU-27, and the United States. Panel (b) illustrates the growth trajectories of AI papers using three-year rolling averages. Panel (c) mirrors the dynamics of

⁶ Note: Fractional counting means that when a publication has authors from multiple countries or regions, the credit is divided proportionally rather than being fully assigned to each.

panel (a) but limits the sample to AI publications with at least ten citations, highlighting more visible contributions. Panel (d) presents the absolute number of AI papers that rank among the top 1% of most-cited publications worldwide. The patterns displayed are specific to Agronomy and should not be generalized to the broader scientific landscape.

Figure 12 depicts the evolution of AI/ML publications in agronomy for the USA, China and the EU27, focusing on both output and impact. Panel (a) shows that China has experienced a dramatic rise, especially after 2016, surpassing both Western regions and reaching close to 8,000 fractional publications by 2023. The EU27 follows a steady, moderate upward trend, while the USA remains relatively stable with only a modest increase in recent years.

Panel B highlights the contrasting growth trajectories among regions. China experienced explosive expansion in the mid-2000s, with growth rates exceeding 30%, followed by a period of deceleration and even negative growth around 2015–2016. After 2018, Chinese growth rebounded strongly. The USA and EU27 maintained steadier, more moderate growth rates. However, since 2021, the USA has exhibited negative growth, signaling a relative stagnation in publication activity.

Panels (c) and (d) highlight differences in research visibility and influence. For publications with at least ten citations (Panel c), China again shows the sharpest growth, peaking around 2019–2020 before tapering off, partly reflecting citation lags for more recent papers. The USA and EU27 contribute at lower levels, with peaks around the same period. In the top 1% of most cited papers (Panel d), China emerges even more clearly as the leading contributor after 2015, while the USA maintains a relatively stable but declining presence and the EU27 plays a more limited role.

Overall, the figure highlights a decisive geographical rebalancing in AI/ML agronomy research. China combines a significant volume of publications with increasing impact, likely supported by sustained investment, data availability, and political prioritization. The United States and the EU27, while still influential, experience more modest expansion, potentially reflecting diversified research portfolios and higher adoption frictions. This model illustrates the global diffusion of AI as a general-purpose technology, where leadership increasingly depends on scale, data, and institutional support.

It is important to emphasize that the primary aim of this thesis is **methodological development** rather than delivering **definitive empirical conclusions**. Accordingly, the results presented in this section serve essentially as illustrative examples that underscore the richness and potential of the constructed datasets. They demonstrate how this resource can support extensive future inquiries into AI diffusion, citation dynamics and textual novelty across scientific domains. Subsequent research can build on this foundation to undertake more focused, substantive investigations within specific disciplines.

5. Final Reflections

5.1 Challenges and Methodological Difficulties

In the process of developing and applying the analytical approach to AI in science diffusion using large-scale bibliometric data, we encountered several challenges. Far from being incidental, these difficulties were formative, shaping both the research design and the analytical competencies acquired throughout the study.

The first set of challenges was technical. Our initial attempt to work with a single aggregated dataset covering all 181 OpenAlex concepts—over 88 million rows—quickly proved infeasible. Local processing

was computationally prohibitive, and the structure of OpenAlex itself added complexity: concepts are hierarchically organized, with identifiers linked to millions of individual works. Navigating this architecture was essential before meaningful analysis could begin. Using BigQuery offered a solution but introduced its own difficulties: queries required careful optimization, billing had to be controlled, and data had to be staged through Google Cloud Storage before download. Once exported, the subdivided concept-level files created additional work, as they often needed to be recombined or harmonized for cross-field comparisons—a process both time-consuming and error-prone without systematic workflows.

To overcome these barriers, we adapted our strategy. Instead of a monolithic dataset, we adopted the concept-by-concept approach described throughout this thesis, which made computations manageable. Pre-filtering and aggregation were pushed upstream into BigQuery to minimize unnecessary exports. Locally, we used memory-efficient processing routines in Python, such as chunked reading, to avoid bottlenecks. Crucially, we emphasized reproducibility: every query, script, and intermediate file was documented to ensure that analyses could be retraced and replicated.

The second set of challenges was conceptual and methodological. In this regard, some core design elements were inherited rather than developed entirely within the scope of this thesis. For instance, the identification of AI-related research was based on a pre-curated list of OpenAlex concepts. While this provided a coherent and standardized entry point, it also introduced potential limitations: the list may be overly broad in some areas, risking the inclusion of publications only loosely connected to AI, while being too narrow in others, potentially omitting relevant work. Similarly, novelty was operationalized through a set of linguistic indicators (semantic distance, new words, word combinations, and phrases) that were supplied as part of the project framework. These measures offered consistency and comparability but inevitably constrained the ways in which novelty could be defined. There were also coverage limitations in the novelty metrics. While the dataset included over 88 million publications with novelty measures, not all publications in our concept-specific datasets could be matched, leaving some papers without corresponding indicators. This highlights that even large-scale bibliometric resources are incomplete and underscores the need for a reproducible framework that can incorporate updated or expanded metrics in future research.

Finally, another methodological limitation concerns the treatment of concept assignments in OpenAlex. Each publication is linked to concepts with an associated confidence score generated by the OpenAlex algorithm. In this study, we adopted a pragmatic approach by using the binary presence of concepts from the provided list, without incorporating confidence scores into the filtering process. This choice was motivated by considerations of tractability and comparability across fields, but it may have introduced additional noise into the dataset. Future work could refine this aspect by weighting concept assignments by confidence levels or testing alternative thresholds.

5.2 Reflections, Applications and Future Directions

Despite these obstacles, the work carried important strengths. The framework developed is scalable, reproducible, and adaptable: it can analyze AI diffusion across hundreds of fields, but could just as easily be extended to other research topics. By combining cloud-based infrastructures with econometric models, the thesis demonstrates a viable approach to studying transformative technologies at scale.

At the same time, limitations remain. Bibliometric indicators are imperfect proxies: citation counts reflect visibility as much as quality, and novelty measures are sensitive to language conventions. The models used here are based on observational data and cannot establish causality. Results must therefore be read as systematic patterns, not as definitive causal claims.

This study also did not explore why AI's effects vary across disciplines. [Bianchini et al. \(2025\)](#) suggests that the structure of field-specific knowledge—how fragmented or interconnected it is—may moderate AI's influence on scientific discovery, with more “rough” knowledge spaces presenting greater potential for transformative impact. Investigating such structural moderators represents a promising avenue for future research.

Nonetheless, the framework has significant applications. Policymakers and funding agencies could use it to monitor the spread of AI/ML across disciplines and regions, identify emerging hubs of activity, and track areas where adoption lags. Researchers can use it to benchmark their fields, while managers of research institutions may find it useful for resource allocation or strategic planning.

This thesis also opens future research directions. The approach could be extended to patent data or collaboration networks, linking publications to technological innovation. It could support longitudinal policy evaluations, testing whether initiatives to promote AI adoption produce measurable effects. Analyses could also be refined at regional or institutional levels, identifying universities or labs driving diffusion. Finally, complementing bibliometric models with qualitative methods such as case studies or expert surveys would deepen interpretation and contextualize quantitative patterns.

Through this process, we also gained skills and competencies that extend beyond the immediate scope of the thesis. Technically, we became proficient in BigQuery, SQL, Python, and cloud workflows for handling massive bibliometric datasets. Methodologically, we learned how to bridge abstract constructs such as “novelty” or “diffusion” with operational measures. Professionally, we strengthened our capacity to adapt in the face of uncertainty, ensure reproducibility, and collaborate effectively in the design of large-scale projects.

In sum, this thesis was as much about the process as the results. By addressing issues of scale, structure, and definition, we built a reproducible framework for studying the diffusion of AI across science. While its findings remain exploratory, the methodological contribution is durable: it provides a foundation for future work at the intersection of data science, bibliometrics, and science policy.

6. Conclusion

Artificial intelligence is no longer confined to specialized domains but increasingly permeates the entire landscape of scientific inquiry, reshaping research across a remarkably broad spectrum of disciplines. From traditional fields like Engineering and Physics to areas as diverse as Agronomy, Fishery, horticulture, and Zoology, AI is now virtually ubiquitous, reflecting its widespread diffusion and transformative potential. The observed trajectory shows a strong and nearly uninterrupted rise in AI-linked publications across the studied fields. While these publications generally exhibit higher scholarly impact, their contributions to novelty are more heterogeneous: AI frequently fosters conceptual recombination, connecting distant areas of knowledge, but does not consistently generate new linguistic or textual markers of innovation. This positions AI as a powerful catalyst for innovation and discovery in science, even as its effects vary across disciplines.

It is worth emphasizing that the focus of this thesis is methodological: the goal is to provide a pilot framework for studying AI diffusion, rather than to draw definitive empirical conclusions. By centering on methods over exhaustive empirical analysis, the work demonstrates both the potential and the challenges of large-scale bibliometric studies, while noting that certain aspects—such as the factors driving heterogeneous effects on novelty or the broader diffusion across regions—were only examined in limited scope (e.g., through a single field-level case study).

All that being said, these findings carry important implications for policy. Some disciplines may be underutilizing AI, indicating opportunities for targeted funding, infrastructure improvements, and specialized training programs. At the same time, widespread AI adoption raises governance challenges: model opacity, biases, and the need for responsible-use frameworks require careful oversight. While overarching principles such as reliability, transparency, and accountability are essential, practical implementation will likely need to be adapted to specific disciplinary and institutional contexts.

Finally, the diffusion of AI invites reflection on the human dimensions of scientific practice. AI excels at processing complex data and identifying patterns, yet it lacks the creativity, intuition, and vision that drive transformative scientific discoveries. Human researchers remain essential for asking the right questions, making conceptual leaps, and navigating uncertainty. In this sense, AI functions as a complement to human intelligence—extending scientific capabilities while highlighting the unique and irreplaceable qualities of human inquiry.

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8. Appendix

8.1 Appendix A| Scientific concepts considered in the study

Openalex ID	Concept Name ⁷	Publication Counts	Share (%)
https://openalex.org/c24890656	Acoustics	2 049 947	26.64%
https://openalex.org/c178802073	Aeronautics	338 209	13.72%
https://openalex.org/c146978453	Aerospace engineering	1 582 882	26.82%
https://openalex.org/c88463610	Agricultural engineering	142 058	14.71%
https://openalex.org/c37621935	Agricultural science	229 333	4.31%
https://openalex.org/c54286561	Agroforestry	363 067	3.48%
https://openalex.org/c6557445	Agronomy	1 725 058	19.5%
https://openalex.org/c11413529	Algorithm	3 829 230	46.81%
https://openalex.org/c105702510	Anatomy	1 331 242	5.11%
https://openalex.org/c16685009	Andrology	290 474	0.51%
https://openalex.org/c42219234	Anesthesia	1 104 050	1.01%
https://openalex.org/c140793950	Animal science	672 789	1.86%
https://openalex.org/c28826006	Applied mathematics	990 144	40.75%
https://openalex.org/c170154142	Architectural engineering	422 355	17.38%
https://openalex.org/c94375191	Arithmetic	327 373	29.0%
https://openalex.org/c87355193	Astrobiology	466 178	3.34%
https://openalex.org/c1276947	Astronomy	1 556 934	15.67%
https://openalex.org/c44870925	Astrophysics	762 684	8.25%
https://openalex.org/c91586092	Atmospheric sciences	457 738	2.98%
https://openalex.org/c184779094	Atomic physics	639 385	3.12%
https://openalex.org/c548259974	Audiology	424 692	18.46%
https://openalex.org/c171146098	Automotive engineering	722 135	27.34%
https://openalex.org/c183696295	Biochemical engineering	164 956	10.08%

⁷ These concepts are Level 1, representing broad scientific domains of AI application.

https://openalex.org/c55493867	Biochemistry	8 396 175	7.74%
https://openalex.org/c60644358	Bioinformatics	478 005	8.13%
https://openalex.org/c186060115	Biological system	169 346	41.64%
https://openalex.org/c136229726	Biomedical engineering	441 986	12.3%
https://openalex.org/c12554922	Biophysics	543 605	1.97%
https://openalex.org/c150903083	Biotechnology	710 406	4.61%
https://openalex.org/c59822182	Botany	2 718 588	9.48%
https://openalex.org/c502942594	Cancer research	1 622 874	0.78%
https://openalex.org/c164705383	Cardiology	1 587 237	3.55%
https://openalex.org/c58640448	Cartography	941 946	21.09%
https://openalex.org/c95444343	Cell biology	2 315 777	0.87%
https://openalex.org/c42360764	Chemical engineering	2 641 911	2.47%
https://openalex.org/c159467904	Chemical physics	395 369	3.53%
https://openalex.org/c43617362	Chromatography	2 775 940	8.05%
https://openalex.org/c147176958	Civil engineering	666 524	15.07%
https://openalex.org/c74650414	Classical mechanics	983 246	25.38%
https://openalex.org/c49204034	Climatology	579 850	6.95%
https://openalex.org/c70410870	Clinical psychology	1 169 780	6.99%
https://openalex.org/c21951064	Combinatorial chemistry	635 378	1.3%
https://openalex.org/c114614502	Combinatorics	1 752 770	28.14%
https://openalex.org/c159985019	Composite material	6 472 276	6.6%
https://openalex.org/c70721500	Computational biology	938 255	10.86%
https://openalex.org/c147597530	Computational chemistry	458 068	4.78%
https://openalex.org/c30475298	Computational physics	202 014	9.17%
https://openalex.org/c26873012	Condensed matter physics	1 159 405	5.74%
https://openalex.org/c107053488	Construction Engineering	136 394	14.9%

https://openalex.org/c133731056	Control engineering	549 653	84.58%
https://openalex.org/c8010536	Crystallography	1 225 206	3.02%
https://openalex.org/c199343813	Dentistry	754 183	5.99%
https://openalex.org/c16005928	Dermatology	604 381	1.37%
https://openalex.org/c47768531	Development economics	357 753	4.34%
https://openalex.org/c118615104	Discrete mathematics	659 700	24.89%
https://openalex.org/c120314980	Distributed computing	678 475	32.35%
https://openalex.org/c1965285	Earth science	147 194	3.34%
https://openalex.org/c18903297	Ecology	5 998 498	8.61%
https://openalex.org/c119599485	Electrical engineering	3 831 992	27.02%
https://openalex.org/c24326235	Electronic engineering	1 354 419	31.7%
https://openalex.org/c194828623	Emergency medicine	483 945	2.23%
https://openalex.org/c134018914	Endocrinology	2 740 229	1.38%
https://openalex.org/c199639397	Engineering drawing	249 584	26.3%
https://openalex.org/c55587333	Engineering ethics	622 563	13.46%
https://openalex.org/c110354214	Engineering management	421 766	15.67%
https://openalex.org/c61696701	Engineering physics	267 703	3.79%
https://openalex.org/c107872376	Environmental chemistry	874 209	2.01%
https://openalex.org/c87717796	Environmental engineering	774 839	6.14%
https://openalex.org/c99454951	Environmental health	2 915 234	4.56%
https://openalex.org/c91375879	Environmental planning	697 620	6.88%
https://openalex.org/c526734887	Environmental protection	242 319	3.4%
https://openalex.org/c107826830	Environmental resource management	664 203	10.17%
https://openalex.org/c78458016	Evolutionary biology	1 049 455	26.33%
https://openalex.org/c512399662	Family medicine	1 321 353	3.62%
https://openalex.org/c505870484	Fishery	791 374	3.83%

https://openalex.org/c31903555	Food science	1 731 993	2.38%
https://openalex.org/c77595967	Forensic engineering	376 568	5.66%
https://openalex.org/c97137747	Forestry	401 315	5.71%
https://openalex.org/c90924648	Gastroenterology	1 206 750	0.49%
https://openalex.org/c61434518	General surgery	468 803	1.82%
https://openalex.org/c54355233	Genetics	6 493 934	4.7%
https://openalex.org/c17409809	Geochemistry	488 536	3.58%
https://openalex.org/c13280743	Geodesy	597 134	33.44%
https://openalex.org/c2524010	Geometry	4 731 727	27.97%
https://openalex.org/c114793014	Geomorphology	662 563	8.78%
https://openalex.org/c8058405	Geophysics	289 016	7.64%
https://openalex.org/c187320778	Geotechnical engineering	1 618 408	10.05%
https://openalex.org/c74909509	Gerontology	875 747	4.39%
https://openalex.org/c29456083	Gynecology	506 151	1.17%
https://openalex.org/c144027150	Horticulture	1 068 867	2.72%
https://openalex.org/c203014093	Immunology	2 860 419	1.17%
https://openalex.org/c13736549	Industrial engineering	95 924	44.89%
https://openalex.org/c179104552	Inorganic chemistry	1 099 742	0.58%
https://openalex.org/c177713679	Intensive care medicine	1 131 030	2.77%
https://openalex.org/c126322002	Internal medicine	12 850 520	2.79%
https://openalex.org/c41895202	Linguistics	5 391 248	23.86%
https://openalex.org/c117671659	Manufacturing engineering	278 875	18.97%
https://openalex.org/c199104240	Marine engineering	296 363	21.07%
https://openalex.org/c134306372	Mathematical analysis	4 284 884	30.97%
https://openalex.org/c126255220	Mathematical optimization	1 299 690	50.81%
https://openalex.org/c37914503	Mathematical physics	494 574	15.92%

https://openalex.org/c78519656	Mechanical engineering	4 230 789	19.3%
https://openalex.org/c57879066	Mechanics	1 883 870	10.84%
https://openalex.org/c509550671	Medical education	1 402 508	10.38%
https://openalex.org/c545542383	Medical emergency	597 848	4.88%
https://openalex.org/c19527891	Medical physics	274 768	17.24%
https://openalex.org/c155647269	Medicinal chemistry	336 576	0.32%
https://openalex.org/c191897082	Metallurgy	3 343 662	3.58%
https://openalex.org/c153294291	Meteorology	2 081 037	11.41%
https://openalex.org/c89423630	Microbiology	1 401 674	1.33%
https://openalex.org/c199289684	Mineralogy	492 750	2.7%
https://openalex.org/c16674752	Mining engineering	204 183	7.54%
https://openalex.org/c153911025	Molecular biology	919 045	0.37%
https://openalex.org/c41999313	Molecular physics	136 402	2.89%
https://openalex.org/c171250308	Nanotechnology	3 454 654	3.8%
https://openalex.org/c169760540	Neuroscience	2 813 925	19.62%
https://openalex.org/c13965031	Nuclear chemistry	701 405	0.28%
https://openalex.org/c116915560	Nuclear engineering	340 708	4.41%
https://openalex.org/c46141821	Nuclear magnetic resonance	289 922	10.16%
https://openalex.org/c2989005	Nuclear medicine	548 474	14.37%
https://openalex.org/c185544564	Nuclear physics	1 001 102	4.31%
https://openalex.org/c159110408	Nursing	2 001 308	5.82%
https://openalex.org/c131872663	Obstetrics	524 862	1.12%
https://openalex.org/c111368507	Oceanography	1 875 645	8.09%
https://openalex.org/c143998085	Oncology	1 245 149	1.73%
https://openalex.org/c21547014	Operations management	932 209	23.6%
https://openalex.org/c42475967	Operations research	618 717	35.22%
https://openalex.org/c118487528	Ophthalmology	474 760	7.37%
https://openalex.org/c120665830	Optics	5 228 649	14.62%

https://openalex.org/c49040817	Optoelectronics	2 758 615	5.02%
https://openalex.org/c119767625	Optometry	198 576	12.97%
https://openalex.org/c178790620	Organic chemistry	8 132 662	3.15%
https://openalex.org/c29694066	Orthodontics	410 492	6.25%
https://openalex.org/c151730666	Paleontology	3 921 940	17.05%
https://openalex.org/c109214941	Particle physics	360 845	4.96%
https://openalex.org/c142724271	Pathology	5 902 577	5.51%
https://openalex.org/c187212893	Pediatrics	1 160 994	1.04%
https://openalex.org/c78762247	Petroleum engineering	375 590	9.77%
https://openalex.org/c5900021	Petrology	175 130	4.58%
https://openalex.org/c98274493	Pharmacology	1 660 415	1.29%
https://openalex.org/c75473681	Photochemistry	478 169	0.56%
https://openalex.org/c147789679	Physical chemistry	2 109 692	3.15%
https://openalex.org/c100970517	Physical geography	350 096	6.81%
https://openalex.org/c99508421	Physical medicine and rehabilitation	733 230	12.43%
https://openalex.org/c1862650	Physical therapy	1 390 561	4.92%
https://openalex.org/c42407357	Physiology	309 369	4.06%
https://openalex.org/c188027245	Polymer chemistry	704 250	6.1%
https://openalex.org/c126348684	Polymer science	127 213	2.14%
https://openalex.org/c21880701	Process engineering	463 291	11.72%
https://openalex.org/c195094911	Process management	674 154	19.09%
https://openalex.org/c118552586	Psychiatry	3 019 430	6.57%
https://openalex.org/c528095902	Pulp and paper industry	435 248	1.79%
https://openalex.org/c202444582	Pure mathematics	2 233 266	25.07%
https://openalex.org/c3079626	Quantum electrodynamics	127 270	3.88%
https://openalex.org/c62520636	Quantum mechanics	9 631 737	22.08%
https://openalex.org/c177322064	Radiochemistry	118 190	1.68%

https://openalex.org/c126838900	Radiology	2 068 334	12.97%
https://openalex.org/c200601418	Reliability engineering	502 409	31.39%
https://openalex.org/c62649853	Remote sensing	967 022	29.75%
https://openalex.org/c112930515	Risk analysis (engineering)	580 606	23.82%
https://openalex.org/c165205528	Seismology	578 874	25.53%
https://openalex.org/c44154836	Simulation	589 979	49.31%
https://openalex.org/c159390177	Soil science	590 314	5.26%
https://openalex.org/c121864883	Statistical physics	620 624	24.42%
https://openalex.org/c105795698	Statistics	5 007 039	35.84%
https://openalex.org/c71240020	Stereochemistry	835 094	1.51%
https://openalex.org/c66938386	Structural engineering	1 996 524	19.17%
https://openalex.org/c141071460	Surgery	4 596 944	2.46%
https://openalex.org/c201995342	Systems engineering	750 199	39.22%
https://openalex.org/c76155785	Telecommunications	3 613 224	28.44%
https://openalex.org/c33332235	Theoretical physics	371 562	7.42%
https://openalex.org/c97355855	Thermodynamics	3 375 915	12.82%
https://openalex.org/c33070731	Toxicology	309 789	1.01%
https://openalex.org/c556039675	Traditional medicine	692 083	1.36%
https://openalex.org/c22212356	Transport engineering	543 475	20.87%
https://openalex.org/c126894567	Urology	364 558	1.4%
https://openalex.org/c42972112	Veterinary medicine	440 773	2.32%
https://openalex.org/c159047783	Virology	1 372 991	2.72%
https://openalex.org/c548081761	Waste management	1 187 845	5.81%
https://openalex.org/c524765639	Water resource management	233 682	11.31%
https://openalex.org/c90856448	Zoology	792 506	2.5%
Total		263 666 637	12.42%

8.2 Appendix B | Query to Select Publications by Scientific Concepts

```
-- =====
-- SQL script: Querying OpenAlex works by concept and exporting
-- Scope: years 2005-2024, language = 'en', type in (article, preprint)
-- =====

DECLARE sql STRING;
DECLARE rows_count INT64;

CREATE SCHEMA IF NOT EXISTS project.dataset
OPTIONS(location = 'US');

-- Step 1: Define target concepts (examples only)
CREATE TEMP TABLE concept_targets AS
WITH inlined AS (
  SELECT s.concept_id, s.concept_name, off AS ord
  FROM UNNEST([
    STRUCT('c147597530' AS concept_id, 'Computational chemistry' AS concept_name),
    STRUCT('c62520636' AS concept_id, 'Quantum mechanics' AS concept_name),
    STRUCT('c169760540' AS concept_id, 'Neuroscience' AS concept_name),
    STRUCT('c143998085' AS concept_id, 'Oncology' AS concept_name),
    -- ... additional concepts included in the full query ...
  ]) AS s WITH OFFSET off
)
SELECT
  REGEXP_REPLACE(INITCAP(concept_name), r'^[A-Za-z0-9]+', '_') AS table_name,
  concept_id,
  concept_name,
  REGEXP_REPLACE(concept_name, r'[/\\]+', '-') AS folder,
  ord
FROM inlined
ORDER BY ord;

-- Step 2: Log table
CREATE TEMP TABLE run_log (
  ord INT64,
  table_name STRING,
  concept_name STRING,
  concept_id STRING,
  rows_count INT64,
  exported BOOL
);

-- Step 3: Loop through concepts
FOR rec IN (
  SELECT table_name, concept_id, concept_name, folder, ord
  FROM concept_targets
  ORDER BY ord
)
DO
  -- Count publications for this concept
  EXECUTE IMMEDIATE '''
    SELECT COUNT(1)
```

```

FROM openalex.works AS w
WHERE EXISTS (
  SELECT 1
  FROM UNNEST(IFNULL(w.concepts, [])) AS c
  WHERE LOWER(c.id) IN (CONCAT('https://openalex.org/', LOWER(@concept_id)))
    OR ENDS_WITH(LOWER(c.id), LOWER(@concept_id))
)
AND w.publication_year BETWEEN 2005 AND 2024
AND LOWER(COALESCE(w.language, '')) = 'en'
AND LOWER(COALESCE(w.type, '')) IN ('article','preprint')
''' INTO rows_count
USING rec.concept_id AS concept_id;

IF rows_count > 0 THEN
  -- Create a concept-specific table
  SET sql = '''
CREATE OR REPLACE TABLE project.dataset.{TABLE} AS
SELECT
  @concept_name          AS matched_concept_name,
  @concept_id            AS matched_concept_id,
  w.id                   AS pub_id,
  w.title,
  w.publication_year,
  w.type,
  w.authors_count,
  w.language,
  w.cited_by_count,
  ARRAY_LENGTH(IFNULL(w.referenced_works, [])) AS reference_count,
  (
    SELECT STRING_AGG(DISTINCT inst.country_code)
    FROM UNNEST(IFNULL(w.authorships, [])) AS a,
         UNNEST(IFNULL(a.institutions, [])) AS inst
    WHERE inst.country_code IS NOT NULL
  ) AS countries,
  (
    SELECT STRING_AGG(
      CONCAT(c.display_name, ' [' , c.id, ']' (level ' , CAST(c.level AS
STRING),
      ' , score ' , CAST(ROUND(c.score, 2) AS STRING), ' ')),
      ';'
    )
  ) AS concepts
FROM UNNEST(IFNULL(w.concepts, [])) AS c
) AS concepts
FROM openalex.works AS w
WHERE EXISTS (
  SELECT 1
  FROM UNNEST(IFNULL(w.concepts, [])) AS c
  WHERE LOWER(c.id) IN (CONCAT('https://openalex.org/', LOWER(@concept_id)))
    OR ENDS_WITH(LOWER(c.id), LOWER(@concept_id))
)
AND w.publication_year BETWEEN 2005 AND 2024
AND LOWER(COALESCE(w.language, '')) = 'en'
AND LOWER(COALESCE(w.type, '')) IN ('article','preprint')
''';
SET sql = REPLACE(sql, '{TABLE}', rec.table_name);

```

```
EXECUTE IMMEDIATE sql USING
  rec.concept_id AS concept_id,
  rec.concept_name AS concept_name;
```

8.3 Appendix C | SQL Command for Exporting to Cloud Storage

```
-- Export data to bucket
SET sql = ''
EXPORT DATA OPTIONS (
  uri='gs://bucket-name/' || @folder || '/' || @folder || '/*.csv.gz',
  format='CSV',
  compression='GZIP',
  overwrite=true,
  header=true,
  field_delimiter=', '
)
AS SELECT * FROM project.dataset.{TABLE}
  '';
SET sql = REPLACE(sql, '{TABLE}', rec.table_name);

EXECUTE IMMEDIATE sql USING rec.folder AS folder;

INSERT INTO run_log VALUES (rec.ord, rec.table_name, rec.concept_name,
rec.concept_id, rows_count, TRUE);
ELSE
  INSERT INTO run_log VALUES (rec.ord, rec.table_name, rec.concept_name,
rec.concept_id, rows_count, FALSE);
END IF;
END FOR;

-- Step 4: Show log
SELECT ord, concept_name, concept_id, rows_count, exported
FROM run_log
ORDER BY ord;
```

8.4 Appendix D| Script for Generating Summary Table of Descriptive and Econometric Result

```
# PYTHON SCRIPT FOR CONCEPT ANALYSIS
# Appendix: data preparation, descriptive statistics, econometric models
# -----

import pandas as pd
import re
from pathlib import Path
import numpy as np
import statsmodels.formula.api as smf
import json

# -----
# PARAMETERS
# -----
folder_path = r"<path_to_csv_folder>"
output_file = "concepts_stats.csv"
json_file = r"<path_to_concept_json>"

# -----
# Load concept IDs
# -----
with open(json_file, "r", encoding="utf-8") as f:
    concepts_data = json.load(f)

concept_ids = {c["id"] for c in concepts_data}
print(f"{len(concept_ids)} concept IDs loaded")

pattern = re.compile(r'C\d+')

def match_concept(concept_str):
    """Return 1 if any concept ID in the string matches our list"""
    if pd.isna(concept_str):
        return 0
    ids = pattern.findall(concept_str)
    return int(any(cid in concept_ids for cid in ids))

# -----
# UTILITY FUNCTIONS
# -----
def get_concept_name_from_file(file_path):
    return file_path.stem.replace("_", " ").title()

def collab(countries):
    """Return 1 if multiple countries are involved"""
    if pd.isna(countries):
        return 0
```

```

    uniq = set([c.strip() for c in countries.split(",") if c.strip()])
    return int(len(uniq) > 1)

def significance_stars(p):
    """Return significance stars based on p-value"""
    if pd.isna(p):
        return ""
    elif p < 0.01:
        return "***"
    elif p < 0.05:
        return "**"
    elif p < 0.1:
        return "*"
    else:
        return ""

# -----
# DATA PREPARATION FOR ECONOMETRIC MODELS
# -----
def prepare_data(df):
    current_year = 2025
    df = df.copy()
    df["age"] = np.maximum(1, current_year - df["publication_year"].astype(int))
    df["tc_w"] = df["cited_by_count"] / df["age"]

    df = df[
        (df["tc_w"] <= 1000) &
        (df["authors_count"] <= 100) &
        (df["reference_count"] <= 1000)
    ].dropna(subset=["tc_w", "authors_count", "reference_count"])

    df = df.groupby("publication_year", group_keys=False).apply(
        lambda g: g.assign(
            most_cited_10=(g["tc_w"] > g["tc_w"].quantile(0.90)).astype(int),
            most_cited_5=(g["tc_w"] > g["tc_w"].quantile(0.95)).astype(int),
            most_cited_1=(g["tc_w"] > g["tc_w"].quantile(0.99)).astype(int),
        )
    )

    for col in ["authors_count", "reference_count", "tc_w"]:
        df[f"{col}_log"] = np.log1p(df[col])

    df["intern_collab"] = df["countries"].map(collab)
    df["survey"] =
df["title"].fillna("").str.contains(r"\b(survey|overview|review)\b",
case=False).astype(int)
    df["is_concept"] = df["concepts"].map(match_concept)

    return df

# -----
# FORMAT RESULTS FOR TABLE
# -----
def format_results_for_table(df_prep):
    df_table = pd.DataFrame(columns=["Model", "Coef_with_signif", "Std. Err."])

```



```

try:
    mod1 = smf.ols(
        "tc_w_log ~ is_concept + authors_count_log + reference_count_log +
intern_collab + survey + C(publication_year)",
        data=df_prep
    ).fit()
    coef = mod1.params.get("is_concept", np.nan)
    se = mod1.bse.get("is_concept", np.nan)
    pval = mod1.pvalues.get("is_concept", np.nan)
    df_table.loc[len(df_table)] = [
        "Weighted Nb. Citations (OLS)",
        f"{coef:.3f}{significance_stars(pval)}",
        f"{se:.3f}"
    ]
except Exception:
    df_table.loc[len(df_table)] = ["Weighted Nb. Citations (OLS)", np.nan,
np.nan]

for name, dep in {
    "Top Cited (10%)": "most_cited_10",
    "Top Cited (5%)": "most_cited_5",
    "Top Cited (1%)": "most_cited_1"
}.items():
    try:
        mod = smf.logit(
            f"{dep} ~ is_concept + authors_count_log + reference_count_log +
intern_collab + survey + C(publication_year)",
            data=df_prep
        ).fit(dis=0)
        coef = mod.params.get("is_concept", np.nan)
        se = mod.bse.get("is_concept", np.nan)
        pval = mod.pvalues.get("is_concept", np.nan)
        df_table.loc[len(df_table)] = [
            f"{name} (Logit)",
            f"{coef:.3f}{significance_stars(pval)}",
            f"{se:.3f}"
        ]
    except Exception:
        df_table.loc[len(df_table)] = [f"{name} (Logit)", np.nan, np.nan]

return df_table

# -----
# PROCESS FILES
# -----
def process_one_file(file_path, concept):
    df = pd.read_csv(file_path)
    df["is_aiml_match"] = df["concepts"].map(match_aiml)

    years = range(2005, 2025)
    results = {"concept": concept}
    results["No of publication"] = len(df)
    results["No of AI publication"] = df["is_aiml_match"].sum()
    results["share of AI papers"] = df["is_aiml_match"].mean()

```

```

    for year in years:
        df_year = df[df["publication_year"] == year]
        results[f"No of publication {year}"] = len(df_year)
        results[f"No of AI publication {year}"] = df_year["is_aiml_match"].sum()
        results[f"share of AI papers {year}"] = df_year["is_aiml_match"].mean() if
len(df_year) > 0 else 0

    # Partie économétrique
    df_prep = prepare_data(df)
    econ_table = format_results_for_table(df_prep)

    # On ajoute les coefficients au dictionnaire
    for _, row in econ_table.iterrows():
        results[row["Model"]] = row["Coef_with_signif"]
        results[row["Model"] + " (SE)"] = row["Std. Err."]

    return pd.DataFrame([results])

def process_all_files(folder_path, output_file):
    results = []
    folder = Path(folder_path)
    csv_files = list(folder.glob("*.csv"))

    for file_path in csv_files:
        concept_name = get_concept_name_from_file(file_path)
        print(f"Processing {concept_name}...")
        stats_df = process_one_file(file_path, concept_name)
        results.append(stats_df)

    final_df = pd.concat(results, ignore_index=True)
    final_df.to_csv(output_file, index=False)
    print(f"Résultats enregistrés dans {output_file}")
    return final_df

def format_results(df):
    df = df.copy()
    for col in df.columns:
        if col.startswith("share of AI papers"):
            df[col] = (df[col] * 100).round(2).astype(str) + "%"
        elif col.startswith("No of publication") or col.startswith("No of AI
publication"):
            df[col] = df[col].apply(lambda x: f"{int(x):,}".replace(",", " "))
    return df

```